Security Risks and the Screening of Investments*

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Abstract

Governments are increasingly restricting foreign investments on national security grounds, yet we lack a clear framework for how governments manage their exposure to foreign firms when geopolitical conditions can deteriorate abruptly. This paper develops a dynamic model in which a host government values foreign investment but faces the risk that some firms are adversarial: during "peace," an adversarial firm can quietly build latent capabilities (e.g., data access, supply-chain dependence) that become harmful if relations turn hostile. This paper characterizes the government's optimal exposure path and the firm's incentives to build. When geopolitical risk is persistent so that hostility can arrive at any time, the government may initially restrict exposure and later liberalize even though the risk of hostility remains unchanged and the firm's type remains uncertain. The ordering changes when risk is anticipated to begin only after an initial period of peace. The government, then, may grant full exposure, retrench before the risky period begins, and liberalize again if peace continues. When hostility occurs at a known date, the final liberalization phase disappears.

Keywords: Dynamic Screening, Information Asymmetry, Geopolitical Risk, Foreign Investment, National Security, Economic Statecraft.

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1 Introduction

Governments across the world are increasingly regulating foreign acquisitions, equity stakes, and other forms of foreign investments on national security grounds. Investment screening mechanisms—regulatory procedures that allow governments to review, condition, or block foreign investments on national security grounds—are a central instrument in this policy shift. Some well known examples of such government intervention are the blocking of the takeover of Aixtron, a German semiconductor equipment manufacturer, by a Chinese firm,[1] and the restrictions on Huawei imposed by the US government.[2] These interventions raise the central question of this paper: How should a government regulate a foreign firm's access over time when that access is economically valuable during peace but may enable the firm to impose harm if geopolitical relations deteriorate?

At first glance, answering this question might seem that screening requires balancing the economic benefits of foreign investment against its potential security costs. Restriction is economically costly: in India, restrictions on Huawei imposed costs on domestic telecommunications firms,[3] while Aixtron needed capital and warned that the failed acquisition would generate economic losses. Granting access and exposure preserves these benefits, but it also allows an adversarial firm to accumulate capacity that can be used if hostility arises. The government would, therefore, compare the marginal economic benefit of exposure with the security cost of the additional capacity it enables. This, however, omits an important informational effect: the activity required to build capacity also creates a risk of detection.

The key insight of this paper is that investment screening is a dynamic problem where the government manages its exposure to foreign firm over time. Each unit of exposure generates

[1] Reuters, "China's Fujian Drops Aixtron Bid After Obama Blocks Deal," December 8, 2016, https://www.reuters.com/article/us-aixtron-m-a-fujian/chinas-fujian-drops-aixtron-bid-after-obama-blocks-deal-idUSKBN13X16H.

[2] Jill C. Gallagher, *U.S. Restrictions on Huawei Technologies: National Security, Foreign Policy, and Economic Interests*, Congressional Research Service Report R47012, January 10, 2022, https://crsreports.congress.gov/product/pdf/R/R47012.

[3] Financial Express, "Ban on equipment from Huawei, ZTE to cost Airtel, Voda-Idea, BSNL dear amid India-China faceoff," June 23, 2020.

an immediate economic benefit and gives an adversarial firm an opportunity to build capacity. The firm chooses how much of that opportunity to exploit, trading off the greater harm it can impose if hostility arises against the higher probability that its activity is detected during peace. Capacity and information arise from the same activity, and the government cannot obtain the latter without enabling the former. A well designed policy, therefore, combines periods of restriction and liberalization as its exposure's economic benefits, the security costs of the capacity it enables, and the information generated through detection as the adversarial firm builds that capacity changes over time.

Interestingly, the ordering of restriction and liberalization depends on whether geopolitical risk is persistent or is anticipated to arise sometime later following a period of peace. Under persistent risk, the government may initially restrict exposure and later liberalize even though the risk of hostility remains unchanged and the firm's type remains uncertain. This liberalization does not occur because the information effect eventually dominates the capacity effect. The adversarial firm builds capacity only while the marginal increase in its hostility payoff exceeds the accompanying loss from detection, so the capacity effect, from the firm's building, always dominates the information effect along the equilibrium path. Information nevertheless narrows the gap between them as exposure accumulates, allowing the economic benefit of exposure to overtake their difference and induce liberalization.

When risk begins only at an anticipated later date, the government may instead grant full exposure early, retrench before the risky period, and liberalize again if peace continues. The initial full exposure arises as the government looks to realize the economic benefits sooner. As the risky period approaches, this economic benefit from offering exposure earlier fades, inducing retrenchment and eventual liberalization as the economic and information effect eventually overcome the capacity effect. If hostility itself occurs at a known date, the final liberalization phase disappears.

These results contribute most directly to the extensive literature on investment screening, which explains the design of screening regulations in the United States and other advanced economies (Canes-Wrone, Mattioli, and Meunier, 2020; Chan and Meunier, 2022; Eichenauer, Dorsch, and Wang, 2021; Kang, 1997). The model builds on three determinants identified by this literature and the broader political economy of foreign investment. The economic gains from exposure capture the interests of domestic firms and other constituencies that benefit from foreign investment (Danzman, 2020; Pandya, 2014). The security costs of firm's capacity building and undermining capture concerns that foreign access to sensitive assets and network may create capabilities that can later be used for coercion (Bauerle Danzman, 2021; Farrell and Newman, 2019). Finally, the firm's hidden type captures the difficulty of distinguishing commercially independent firms from those susceptible to home-state pressures (Davis, Fuchs, and Johnson, 2019; Pearson, Rithmire, and Tsai, 2022). In the model, this uncertainty makes detection and continued non-detection informative. Building on these determinants, the paper focuses on how the access granted and the concurrent screening should evolve over time.

The mechanism in this model builds on a broader literature on models of strategic behavior under imperfect information and hidden actions. Granting exposure produces economic benefits as well as future security costs when an adversarial firm uses this exposure to build capacity. This component of the mechanism builds on models of bargaining and conflict. Private information about actors' types and actions can make choices taken before hostility consequential for later bargaining and conflict (Fearon, 1995; Fey and Ramsay, 2007; Kenkel and Schram, 2024). Work on cooperation and commitment problems further shows how present arrangements can alter future bargaining power, alliance commitments, and deterrence incentives (Benson and Smith, 2023; Bils and Smith, 2025; Powell, 2006; Smith, 2021). Related models demonstrate how investments and policy choices made before a crisis reshape the capabilities, threats, and bargains available afterward (Bas and Coe, 2016; Bils and Spaniel, 2017; Coe and Vaynman, 2020; Debs and Monteiro, 2014). In the context of foreign investment, the problem is therefore not only that the government is uncertain about the firm's preferences. Access granted today can also change what the firm is capable of doing tomorrow.

The second component in this model is the information generated through exposure and the firm's actions. This connects the paper to models of imperfect monitoring and hidden actions. The closest benchmarks are models of dynamic trust and screening. Kolb and Madsen (2023) study how a principal adjusts the stakes of a relationship while screening an agent of uncertain loyalty. Zalke (2026) builds on this principal-agent problem by allowing an adversarial agent to choose an imperfectly observed harm level before the relationship begins, requiring the principal to screen both the agent's loyalty and its damage potential. The hostility-stage analysis in this paper builds on that characterization. In domestic and conflict settings, related models show how actors make choices when monitoring is imperfect and when later political payoffs depend on what has or has not been revealed (Gibilisco and Steinberg, 2023; Hirsch and Kastellec, 2022).

This paper combines the theoretical work with the substantive findings from the literature on investment screening to present a framework explaining how governments manage exposure to foreign firms while allowing them access in their economy. The economic benefit of maintaining this access is central to the broader literature on foreign investment and economic openness. Existing work explains governments' preference for economic openness through domestic interests, democratic institutions, partisanship, financial development, and firms' demand for external financing (Danzman, 2020; Kalinova, Palerm, and Thomsen, 2010; Milner, 1997; Milner and Kubota, 2005; Pandya, 2014; Pinto, 2013; Rajan and Zingales, 2003). Related studies show how political risk, international commitments, and competition for mobile capital affect foreign investment (Büthe and Milner, 2008; Elkins, Guzman, and Simmons, 2006; Jensen, 2008; Li and Resnick, 2003; Wellhausen, 2015).

Foreign firms also become embedded in domestic production networks, creating domestic constituencies with an interest in preserving their access (Johns and Wellhausen, 2016; Pandya, 2016). These studies explain why restricting a firm that is already present in the economy can be costly even when the government's security concerns are genuine. However, whereas much of this literature examines how institutions protect a foreign investor against host-country political risk, this paper considers the reverse problem: how the host government protects itself when an embedded foreign firm may become strategically dangerous.

The source of this danger connects the paper to work on economic statecraft and interdependence. Economic relationships can create asymmetric dependence and political leverage alongside economic gains (Baldwin, 1986; Hirschman, 1980; Keohane and Nye Jr, 1973; Mastanduno, 1998). More recent work shows how states exploit their position within economic networks and how security concerns increasingly shape trade and investment policy (Drezner, Farrell, and Newman, 2021; Farrell and Newman, 2019; Meunier and Nicolaïdis, 2019; Roberts, Choer Moraes, and Ferguson, 2019).

Whether such interdependence creates a security concern depends partly on the geopolitical relationship between the host and source countries (Colaresi and Thompson, 2002; Thompson, 2001). It also depends on the relationship between the foreign firm and its home government: state control can make firms responsive to interstate relations through political connections with their home country government. This can blur the distinction between public and private actors (Davis, Fuchs, and Johnson, 2019; Li, Meng, et al., 2008; Pearson, Rithmire, and Tsai, 2022). These arguments motivate the model's uncertainty about the firm's type and its capacity effect.

Bringing these arguments together produces the paper's substantive contribution. Existing research explains why governments value foreign investment, how economic interdependence can become a source of vulnerability, why some firms attract greater security scrutiny, and why governments create screening institutions. The model connects these elements together and examines how a government can design policy to manage these various effects. Interestingly, the design of the policy depends upon when the government expects geopolitical risks to begin. When these risks are persistent, the government starts cautiously, restricting access and then, along continued non-detection, liberalizes even when the geopolitical risk remains. When the government anticipates geopolitical risk to begin only after a certain time, they may, instead, grant maximum access early on, restricting it as time goes on and eventually liberalize.

The rest of the paper proceeds as follows. Section 2 presents the model. Section 3 presents the analysis. Section 4 analyzes the case when geopolitical situation can worsen suddenly. Section 5 analyzes a setup where the government anticipates an increase in geopolitical tension after a peaceful period. Section 6 turns to the setup in which the onset of hostility is known. The final section concludes with a discussion on the implications for investment screening and security-motivated restrictions on foreign investments and future directions for research.

2 Model

There are two players: a host government (G) and a foreign firm (F). The foreign firm can be of two types $\theta \in \{g, b\}$, where $\theta = g$ denotes a commercial (good) firm and $\theta = b$ an adversarial (bad) firm that may act to undermine the host country.

Time is continuous, $t \geq 0$, and there can be two states of the world: peace (p) or hostility (h). The relationship starts in peace but can turn to hostility due to a geopolitical shock. In the baseline environment, which we call **persistent geopolitical risk**, the shock arrives randomly according to a Poisson process with hazard rate λ. Once the state turns hostile, it is absorbing.

At each time t, the government decides how much access to provide to the firm: it can either terminate the relationship with the firm in which case the firm will be excluded from its economy forever from then on, or, choose to continue the relationship, in which case, the government determines its exposure to the firm. The feasible exposure level depends on the geopolitical state. During peace, the government chooses $x_t^p \in [\phi, 1]$; during hostility, it chooses $x_t^h \in [\phi_h, 1]$, where $0 < \phi_h < \phi < 1$. The exposure x_t^j, where $j \in \{p, h\}$, can be understood as a "normalized exposure index" that captures the effective integration of the foreign firm in the domestic economy. Accordingly, $x_t^j = \phi_j$ and $x_t^j = 1$ denote the feasible minimal and maximal exposure, respectively, compatible with the geopolitical state j. We assume that ϕ_h can be arbitrarily small but positive.[4] Both players discount future payoffs at rate $r > 0$. We assume the government has commitment power within each geopolitical state, but not across states.

During peace, an adversarial firm can use the exposure to build capacity. At time t in peace, it chooses a capacity-building effort $\alpha_t \in [0, 1]$, so its accumulated capacity, C_t, which we denote as capacity stock, evolves according to $\dot{C}_t = \alpha_t x_t$. The commercial firm does not accumulate any capacity. For ease of calculations, we set $\alpha = 0$ for a commercial firm. The capacity building does not hurt the government directly. Thus, its flow payoff during peace is

$$\pi_t^p(x_t) = x_t.$$

Once hostility arrives at date T_h, the capacity stock freezes and remains fixed at C_{T_h}. An adversarial firm can now use this capacity to secretly undermine the government. It chooses an undermining intensity $\beta_t \in [0, 1]$ that inflicts a flow loss of $(1 + KC_{T_h})\beta_t x_t$ on the host government, where $K > 0$. The multiplier $k(C_{T_h}) := 1 + KC_{T_h}$ denotes the effective harm per unit of exposure and undermining. Therefore, the flow payoffs for the government during hostility conditional on the firm being adversarial with capacity stock C_{T_h} and undermining intensity β_t is

$$\pi_t^h(x_t, \beta_t; C_{T_h}) = x_t - (1 + KC_{T_h})\beta_t x_t.$$

We consider a zero-sum game between the goevernment and the adversarial firm during hostility. Therefore, the adversarial firm's hostility flow payoffs are $-\pi_t^h(x_t, \beta_t; C_{T_h})$. The payoffs for the government and the commercial are aligned even during hostility, and therefore, the commercial firm does not undermine. We can, therefore, set $\beta = 0$ for the commercial firm to reflect this.

[4] Allowing ϕ_h to be very small reflects most screening settings where regulators can feasibly constrain foreign firms' effective integration without fully terminating the relationship. In practice, this can be done by limiting the scope of the firms' activities, such as, capping interfaces and data permissions, restricting any expansion of the firm.

Information. The firm's type is private information. The government's belief at time t that the firm is commercial is $q_t \in [0, 1]$, with initial belief $q_0 \in (0, 1)$. During peace, whenever the adversarial firm builds capacity with effort α_t over an interval dt, the government receives a private and unverifiable signal about its type with probability $\mu\alpha_t x_t dt$, where $\mu > 0$ denotes the effectiveness of government's peace-time monitoring. Note that detection is not cumulative in past capacity building. During hostility, each act of undermining is discovered at rate $\gamma\beta_t dt$, where $\gamma > 0$ reflects the quality of monitoring. The government does not observe its payoff until the end of the game.

Contracts and Timing. The government's access schedule and exclusion policy within a geopolitical state $j \in \{p, h\}$ can be denoted by a contract $\mathcal{C}^j = (x^j, T_R^j)$, where $x^j = (x_t^j)_t$ specifies the exposure while access remains active and T_R^j denotes the time when the firm's access is terminated. The relationship begins in peace, when the government offers single peace-time pooling contract, $\mathcal{C}^p$. Upon receiving this contract, an adversarial firm chooses its capacity-building effort. When the geopolitical shock arrives, capacity accumulation ends, the peace-time contract expires, and the government commits to a new contract, $\mathcal{C}^h$, specifying the firm's exposure and termination during hostility. The adversarial firm then chooses its undermining intensity until the relationship is terminated. Termination ends the game, and each player collects their payoff.

Discussion of Assumptions

Unverifiable Signal During Peace. We assume that the signal that the government receives during peace is private and unverifiable. In the context of investment screening, the information that the government gets about latent capacity may originate as classified intelligence or other channels where the government cannot credibly demonstrate this to the firm or any other adjudicating authority. Such a setting where the principal has information that is non-contractible is common across many applications (Aghion, Dewatripont, and Rey, 1994; Baker, Gibbons, and Murphy, 1994; Baliga, Bueno de Mesquita, and Wolitzky, 2020). By contrast, the signal during hostility is generated by an act of undermining for which the government can produce verifiable evidence. Thus, we allow the government to condition the hostility contract on the history of detected undermining.

Pooling Contracts. We restrict the government to a single contract within each geopolitical state. A single pooling peace-time contract rules out a menu of contracts in which the firm's selection reveals its type during peace. This restriction captures the political cost that an adversarial firm may face from its home country if it voluntarily reveals itself to the host government. Publicly observed choices that reveal a player's alignment may have strategic consequences (Bils and Smith, 2025; Smith, 2021). In the context of investment screening, selecting a contract intended for an adversarial firm would constitute an observable act of cooperation with the host government which could result in the firm's home government imposing significant costs on the firm. In particular, foreign firms may depend on their home governments for financing, licenses, or even legal protection which allows the home government to discipline these firms (Davis, Fuchs, and Johnson, 2019; Li, Meng, et al., 2008; Pearson, Rithmire, and Tsai, 2022). The resulting political costs could be sufficiently large that an adversarial firm would not voluntarily choose a contract that reveals its type. A separating menu, therefore, is ineffective during peace.

We impose the same single-contract restriction during hostility. However, this restriction is without loss in the hostility game: the government cannot improve by offering a type-contingent menu in a zero-sum game and may restrict attention to a single stringent pooling contract (Kolb and Madsen, 2023). Thus, this restriction does not constrain the government's optimal hostility contract.

3 Analysis

The government offers a peace-time contract $\mathcal{C}^p \in \mathfrak{C}^p$ at $t = 0$, where $\mathcal{C}^p$ states an exclusion time $T_R^p \in [0, \infty]$ and an exposure path $x^p = (x_t^p)_{t \leq T_R^p}$, with $x_t^p \in [\phi, 1]$, that applies when the relationship remains active during peace. If hostility arrives before T_R^p, $T_h < T_R^p$, the peace-time contract expires at T_h. The government, however, never excludes the firm during peace. The following lemma states this formally.

Lemma 1. *Excluding the firm during peace is strictly suboptimal.*

Before T_h, the payoffs remain aligned and the firm cannot impose security costs. If capacity building is detected, the government can always exclude the firm at the onset of hostility ensuring that it does not incur any loss. Hence, excluding the firm early only sacrifices the peace-time flow payoff and yields no additional security benefit. We can, therefore, restrict attention to contracts with $T_F^p = \infty$, and define the peace time contract $\mathcal{C}^p$ only by its exposure path x^p.

During peace, the firm observes the offered contract $\mathcal{C}^p$, knows its type, and remembers its capacity building choices. Let $\mathcal{H}_t^{F,p}$ denote the set of the firm's possible information histories at time t, while peace and the relationship continue. An adversarial firm's pure capacity-building strategy is $\sigma_t^{F,p} : \mathcal{H}_t^{F,p} \longrightarrow [0,1]$ and its effort at t is $\alpha_t = \sigma_t^{F,p}(h_t^{F,p})$.

For the government, let $\mathcal{H}^{G,h}$ denote the set of the histories which record the peace-time contract, the hostility arrival date, and the government's private detection history. Its hostility-contract strategy is $\sigma^{G,h} : \mathcal{H}^{G,h} \longrightarrow \mathfrak{C}^h$, where $\mathfrak{C}^h$ is the set of hostility contracts. At a history $h^{G,h}$, write $q(h^{G,h})$ for its posterior probability that the firm is commercial and $P^C(h^{G,h})$ for its conditional posterior over capacity if the firm is adversarial. The posterior over effective harm is induced by $k(c) = 1 + Kc$.

A hostility-contract strategy that depends on history only through these posterior beliefs admits the representation $\sigma^{G,h}(h^{G,h}) = \widehat{\sigma}^{G,h}\left(q(h^{G,h}), P^C(h^{G,h})\right)$. We introduce such reduced representations when characterizing the continuation problems below.

During hostility, the firm retains its earlier information and observes the offered hostility contract. It also remembers its own undermining choices. Let $\mathcal{H}_t^{F,h}$ denote its possible information histories while the relationship remains active. An adversarial firm's pure undermining strategy is $\sigma_t^{F,h} : \mathcal{H}_t^{F,h} \longrightarrow [0,1]$. The commercial firm chooses $\alpha_t = \beta_t = 0$.

The equilibrium concept is Perfect Bayesian Equilibrium. The government's contract choices are optimal at the beginning of each geopolitical state, given its beliefs and the firm's continuation strategy. The firm's continuation choices are optimal at every information history, including histories following its own earlier deviations and the offer of any admissible contract. Beliefs are updated according to Bayes' rule wherever possible.wherever possible.

4 Baseline: Persistent Geopolitical Risk

In the baseline, geopolitical shock arrives randomly with hazard rate $\lambda > 0$. We solve the game backward, first characterizing the optimal hostility contract for every belief inherited from peace and then solving the government's peace-time contracting problem using the resulting hostility continuation values.

Hostility Continuation

If capacity building was detected during peace, the government knows that the firm is adversarial and optimally excludes it when hostility begins. We therefore characterize the hostility continuation conditional on no peace-time detection. At T_h, the government has a posterior belief q that the firm is commercial and, conditional on it being adversarial, a posterior $P^C_{T_h}$ over its capacity. To characterize the hostility continuation, we write q for the posterior q_{T_h} at the end of peace and P^C for $P^C_{T_h}$.

Consider first a degenerate belief $P^C = \delta_c$, so that the government, at the beginning of hostility, believes that the adversarial firm has accumulated capacity c during peace. Thus, the government believes the adversarial firm's effective harm to be $k(c) := 1 + Kc$. For a given $(q, k(c))$, the hostility continuation game coincides with the screening problem in Kolb and Madsen (2023). They show that it is without loss to use a stringent contract that excludes the firm upon detected undermining. Furthermore, because the minimal access, ϕ, that the government can restrict the firm to is sufficiently small, the optimal hostility contract never terminates the relationship in the absence of detection.

At the onset of hostility, if the government's belief that the firm is commercial is sufficiently high,

$$q \geq \frac{k(c)-1}{k(c)},$$

it grants full exposure immediately. If instead, $q \leq \frac{k(c)-1}{k(c)}$ at the start of hostilities, the government offers a testing contract: exposure is initially restricted, then rises following continued non-detection, and eventually reaches one. The adversarial firm remains indifferent between undermining with full intensity, $\beta_t = 1$, and not undermining at all $\beta_t = 0$ for all t during hostility. At $q = \frac{k(c)-1}{k(c)}$, the two contracts coincide. Proposition A.6 characterizes the optimal hostility contract as well as the hostility payoffs. Thus, conditional on no peace-time detection, when $P^C = \delta_c$, the government's optimal hostility contract and the corresponding continuation payoff depend only on the posterior belief q, and the effective harm, $k(c)$, which the government believes the adversarial firm possesses.

To see how peace-time capacity accumulation and exposure shape these beliefs at the onset of hostility, fix an exposure path x^p and consider a time t during peace that is reached while the relationship remains active. For any capacity build path α that the adversarial firm chooses, since peace-time detection occurs at the rate $\mu\alpha_t x_t^p$, the probability that an adversarial firm remains undetected through time t is $e^{-\int_0^t \mu\alpha_s x_s^p\, ds} = e^{-\mu C_t(\alpha; x^p)}$. Thus, the probability of detection depends on the path α only through the capacity stock $C_t(\alpha; x^p)$. A commercial firm never builds any capacity. Therefore, if at time t, the government believes that the adversarial firm's capacity stock is $C_t(\alpha; x^p) = c$, then conditional on no detection, her posterior belief that the firm is commercial is

$$q_t = q(c) = \frac{q_0}{q_0 + (1-q_0)e^{-\mu c}}. \tag{1}$$

Equation (1) implies that, conditional on no detection through t, any two capacity-building paths, α^1 and α^2, yielding the same belief over capacity stock, $P^C = \delta_c$, induce the same posterior $q(c)$. If hostility begins at t, the government, therefore, enters the hostility phase with the same belief pair $(q(c), \delta_c)$ under either path and offers the same hostility contract $\mathcal{C}^{h*}(q(c), k(c))$. Therefore, the optimal hostility contract depends only on the posterior beliefs over the capacity stock and not on the build path, α, that induces it.

Under the belief pair $(q(c), \delta_c)$, the hostility contract characterization above implies that the optimal hostility contract grants full exposure if and only if

$$q(c) \geq [1 - q(c)]Kc \quad \Longleftrightarrow \quad Kce^{-\mu c} \leq \frac{q_0}{1 - q_0}, \tag{2}$$

where the equivalence follows from (1) and $k(c) = 1 + Kc$. Let $Y(c) := Kce^{-\mu c}$ denote the excess harm generated by the capacity stock c, weighted by the probability that the firm remains undetected while accumulating that capacity. The function $Y(c)$ is single-peaked in c and attains its unique maximum at $C_Y = 1/\mu$, where $Y(C_Y) = K/\mu e$. The following proposition characterizes the optimal hostility contract when the government believes that the adversarial firm has accumulated capacity c.

Proposition 1. *Suppose at T_h, the government believes that an adversarial firm has capacity stock c. Then, along the no-detection history*

- *if $\frac{K}{\mu e} \leq \frac{q_0}{1-q_0}$, the government grants full exposure immediately for every inherited capacity c.*
- *if $\frac{K}{\mu e} \geq \frac{q_0}{1-q_0}$, there exist unique c^-, c^+, where $c^- \leq C_Y \leq c^+$, satisfying $Y(c^-) = Y(c^+) = \frac{q_0}{1-q_0}$, such that the government grants full exposure when $c \leq c^-$ or $c \geq c^+$, and offers a testing contract when $c \in [c^-, c^+]$.*

The two contracts coincide at $\frac{K}{\mu e} = \frac{q_0}{1-q_0}$ with $c^- = C_Y = c^+$. If the adversarial firm is detected before T_h, the government excludes the firm.

The result in Proposition 1 reflects the two opposing effects of capacity stock c on the government's hostility contract. First, a higher capacity c raises the effective harm $k(c)$ that the government believes an adversarial firm can impose, making a testing contract attractive. Second, an adversarial firm is less likely to remain undetected while accumulating a larger capacity stock. No detection, therefore, becomes stronger evidence that the firm is commercial, raising $q(c)$ and making full exposure more attractive. Together, these effects make the survival weighted excess harm $Y(c)$ single-peaked, increasing up to C_Y, and decreasing thereafter.

Whether the government offers a testing contract during hostility depends on whether the maximum, $Y(C_Y) = K/\mu e$, exceeds the prior odds of the firm being commercial, $q_0/(1-q_0)$. If it does not, the government offers full exposure for every c. If the maximum exceeds this threshold, the government grants full exposure when $c \leq c^-$ or $c \geq c^+$ and offers a testing contract for $c \in [c^-, c^+]$.

The preceding analysis considered the case in which the government's posterior over the adversarial firm's capacity is degenerate, i.e., $P^C = \delta_c$. More generally, the adversarial firm may randomize over capacity-building paths in equilibrium. Fix an exposure path x^p, and let $H_t \in \Delta(\mathbb{R}_+)$ denote the distribution over capacity stock C_t induced by this randomization. Conditional on no detection through t, the government's posterior beliefs over the type and capacity stock is

$$q_t = \frac{q_0}{q_0 + (1-q_0)\int e^{-\mu c}\, dH_t(c)} \quad ; \quad P_t^C(A) = \frac{\int_A e^{-\mu c}\, dH_t(c)}{\int e^{-\mu c}\, dH_t(c)}. \tag{3}$$

Equation (3) shows that, in equilibrium, the government's belief pair (q_t, P_t^C) along the no-detection history, and thus, the hostility contract, depends on the adversarial firm's randomization over capacity-building paths only through the induced capacity distribution, H_t. Moreover, conditional on hostility beginning at t, any two randomization that induce the same H_t generate the same expected continuation payoffs under every hostility contract. Thus, in equilibrium, at any given t along the no-detection history, capacity distribution H_t and the corresponding belief pair (q_t, P_t^C), determines the hostility continuation problem.

The next step is to determine the distribution H_t induced by the adversarial firm's equilibrium strategy and the hostility contract that the government offers in equilibrium.

Capacity Building and Cumulative Exposure

Fix a time t during peace at which the relationship remains active, and let $X_t = \int_0^t x_s^p \, ds$ denote cumulative exposure through t under the exposure path x^p. Since $\dot{C}_s = \alpha_s x_s^p$ and $\alpha_s \in [0,1]$, the adversarial firm can attain any capacity stock $c \in [0, X_t]$ by t. Let $u^h(\mathcal{C}^h, c)$ denote the adversarial firm's hostility payoff under contract $\mathcal{C}^h$ when its capacity is c. If hostility begins at t, a firm with capacity stock c obtains this continuation payoff only if it has remained undetected, which occurs with probability $e^{-\mu c}$. Thus, its expected hostility payoff is $e^{-\mu c} u^h(\mathcal{C}^h, c)$. Because the firm's peace-time flow payoff is fixed by the exposure path and does not depend on its capacity-building, it does not affect the firm's choice.

If the adversarial firm's strategy induces a distribution $H_t \in \Delta([0, X_t])$, its expected hostility payoff under a contract $\mathcal{C}^h$ is

$$U^F(H_t, \mathcal{C}^h; X_t) = \int_0^{X_t} e^{-\mu c} u^h(\mathcal{C}^h, c) \, dH_t(c). \tag{4}$$

The adversarial firm, however, does not choose H_t independently at each possible hostility t. Given an exposure path x^p, its capacity-building strategy induces the entire family $(H_t)_{t \geq 0}$.

We proceed by first solving an auxiliary problem for a fixed cumulative-exposure level X, which is an application of the constrained endogenous-harm screening problem studied in Zalke (2026). [5] Second, we show that the resulting family of capacity distributions can be generated by a single feasible capacity-building strategy satisfying $\dot{C}_t = \alpha_t x_t^p$, and that this strategy is optimal in the firm's dynamic capacity-building problem.

We begin with the auxiliary problem. Fix a cumulative exposure level $X \geq 0$. The adversarial firm privately chooses a capacity $c \in [0, X]$ and may randomize. Denote the resulting distribution over capacity by $H \in \Delta([0, X])$. In equilibrium, conditional on no detection, H induces the government's belief pair $(q(H), P^C(H))$ according to equation (3), following which the government offers a hostility contract $\mathcal{C}^h$ based on those beliefs.

[5] In the endogenous-harm screening problem studied in Zalke (2026), a principal offers a single hostility contract against a distribution of privately known capacities that is selected endogenously by an agent.

An equilibrium of this auxiliary problem consists of a distribution H_X^* and a hostility contract $\mathcal{C}_X^{h*}$ such that: given the beliefs induced by H_X^*, the government's optimal hostility contract, $\mathcal{C}^{h*}$, maximizes $V^h(\mathcal{C}^h; q(H_X^*), P^C(H_X^*))$; and given $\mathcal{C}_X^{h*}$, the adversarial firm's distribution over capacity, H_X^*, maximizes $U^F(H, \mathcal{C}_X^{h*})$.

If the adversarial firm is detected during peace, the government excludes the firm at the onset of hostility. Continuing to keep an adversarial firm in a zero-sum game is weakly suboptimal. [6] Thus, for an equilibrium pair $(H_X^*, \mathcal{C}_X^{h*})$, the government's ex ante hostility payoff when hostility begins after cumulative exposure X is

$$\Psi(X) = \left[q_0 + (1 - q_0) \int_0^X e^{-\mu c} \, dH_X^*(c)\right] V^h\big(\mathcal{C}_X^{h*}; q(H_X^*), P^C(H_X^*)\big) \,, \tag{5}$$

where the term in the bracket is the probability that an adversarial firm is not detected before hostility.

We begin by showing that the firm cannot play a pure strategy. To see this, when $\frac{K}{\mu e} \geq q_0/(1-q_0)$, the government offers a testing contract if it believes $c \in [c^-, c^+]$ (Proposition 1). It is implied from the definitions of C_H and c^+, that $C_Y < C_H < c^+$.[7] Suppose $C_Y < X < C_H$ and the government offers a testing contract calibrated to X believing that is the capcity stock of the adversarial firm. Recall that such a testing contract will keep an adversarial firm with effective harm $k(X)$ indifferent between undermining with full intensity and not undermining at all Proposition A.6. However, under this testing contract, a deviating firm, $c' \in [C_Y, X)$, gets a lower hostility payoff (reduced by the factor c'/X),[8] conditional on reaching hostility undetected. However, it has a higher probability of getting to hostility

[6] We will show that, in equilibrium, the effective harm given the adversarial firm's capital stock, c^*, is strictly positive. Hence, not excluding the firm at the onset of hostility upon detection is strictly suboptimal.

[7] Testing at C_Y implies $w(C_Y) < \mu C_Y = 1$. At c^+, $q(c^+) = Kc^+/(1 + Kc^+)$, so $w(c^+) = \mu c^+ > 1$. Since $w'(c) > 0$, it crosses one uniquely satisfying $C_Y < C_H < c^+$. At equality, $C_H = C_Y = c^+$.

[8] A deviating firm with $c' < X$ can undermine with full intensity, $\beta_t = 1$, giving it a flow payoff of $Kc'x_t^h$ instead of KXx_t^h, while facing the same detection rate during hostility, $\gamma\beta_t \, dt = \gamma \, dt$. Hence, its conditional hostility payoff is c'/X times her payoff from choosing $c = X$.

undetected. This yields a higher ex-ante hostility payoff because

$$\frac{e^{-\mu c'}u^h(\mathcal{C}^{h*},c')}{e^{-\mu X}u^h(\mathcal{C}^{h*},X)} = \frac{e^{-\mu c'}}{e^{-\mu X}}\frac{c'}{X} > 1, \tag{6}$$

making deviation profitable.

If, instead, the government offers a testing contract calibrated to some $c' \in [C_Y, X)$, the firm can deviate to some $c = c' + \delta < X$, where δ is sufficiently small. Since, under this contract, $\beta_t = 0$ is optimal for a firm with effective harm $k(c')$, the deviating firm can follow the same strategy and get,

$$e^{-\mu(c'+\delta)}u^h(\mathcal{C}^{h*},c'+\delta) = e^{-\mu(c'+\delta)}A(c')\frac{q(c')(1+Kc')+K\delta}{r+\gamma} > e^{-\mu c'}u^h(\mathcal{C}^{h*},c'). \tag{7}$$

where the expression for $u^h(\mathcal{C}^{h*},\cdot)$ follows from Proposition A.6. Thus, the deviating firm has a lower probability of not getting detected, however, conditional on not getting detected, it gets a higher hostility payoff yielding a higher ex ante hostility payoff (equation (7)).

Therefore, no hostility testing-contract characterized in Proposition A.6 calibrated to a single capacity can sustain a pure capacity choice. The equilibrium, instead, uses the endogenous-harm screening contract characterized in Zalke (2026), applied to this setup. The following proposition characterizes the firm's equilibrium capacity distribution and the government's corresponding hostility contribution.

Proposition 2 (Equilibrium-fixed X)**.** *Consider the auxiliary problem where the cumulative peace-time exposure is fixed at $X \geq 0$. Define $\underline{c}_X = \min\{X, C_Y\}$ and $\overline{c}_X = \min\{X, C_H\}$ where,*

$$w(c) = \mu q(c)\left(c + \frac{1}{K}\right), \quad C_H = \min\{c \geq C_Y : w(c) \geq 1\}, \quad M(c) = (1+Kc)\big(1-q(c)\big).$$

Then, in equilibrium,

- *the capacity distribution is unique and has the following functional form:*

$$H_X^*(c) = \begin{cases} 0, & c < \underline{c}_X, \\ \dfrac{1 + \left(c - \underline{c}_X\right)K}{M\left(\overline{c}_X\right)}, & \underline{c}_X \leq c < \overline{c}_X, \\ 1, & c \geq \overline{c}_X. \end{cases} \tag{8}$$

- *The government's ex ante hostility payoff is*

$$\Psi(X) = \begin{cases} \dfrac{q_0}{r} - \dfrac{1-q_0}{r+\gamma} K \underline{c}_X e^{-\mu\kappa}, & X \leq c^-, \\ \dfrac{q_0 \gamma}{r(r+\gamma)} e^{-rD(X)}, & X \geq c^-, \end{cases} \tag{9}$$

where c^- is as characterized in Proposition 1, with $c^- = \infty$ when $K/\mu e \leq q_0/(1-q_0)$, and

$$D(X) = \frac{\mu}{2\gamma K}\left[(1 + K\overline{c}_X)^2 - (1 + K\underline{c}_X)^2\right] - \frac{1 + K\overline{c}_X}{\gamma} \log \frac{w(\overline{c}_X)}{\mu \underline{c}_X}.$$

From equation (8), we can see that the firm never chooses capacity stock $c < \underline{c}_X = \min\{X, C_Y\}$ and $c > \overline{c}_X = \min\{X, C_H\}$.[9] The capacity distribution characterized in Proposition 2 reflects the adversarial firm's trade-off between the hostility benefit of greater capacity and the probability of surviving peace-time detection. To see this, suppose first that the government grants full exposure when hostility begins. We can reset the clock to 0 at T_h. Recall that an adversarial firm always undermines with full intensity, $\beta_t = 1$, when offered full exposure, $x_t = 1$ for all $t \geq 0$ (Proposition A.6). Conditional on reaching hostility undetected, an adversarial firm with capacity c then receives the hostility payoff

$$u^h(\mathcal{C}^h, c) = \int_0^\infty e^{-(r+\gamma)t}\left[(1 + Kc)\beta_t x_t - 1\right] dt = \int_0^\infty e^{-(r+\gamma)t}\left[(1 + Kc)\beta_t x_t - 1\right] dt = \frac{Kc}{r+\gamma}.$$

Its expected payoff is, therefore,

$$e^{-\mu c} u^h(\mathcal{C}^h, c) = e^{-\mu c}\frac{Kc}{r+\gamma} = \frac{Y(c)}{r+\gamma}. \tag{10}$$

[9] It follows from (6) that the firm has a profitable deviation to C_Y whenever the hostility contract is more stringent (calibrated to $c > C_Y$. Equation (7) shows the profitable upper deviation when the contract is lenient and upper deviation is feasible. C_H is the capacity stock for which there is no upper deviation, i.e., $\frac{\partial}{\partial c} e^{-\mu c} u^h(\mathcal{C}^{h*}, c) \mid_{C_H} = 0$, when $\mathcal{C}^{h*}$ is calibrated for C_H.

Recall from our discussion preceding Proposition 1 that the function $Y(c) = Kce^{-\mu c}$ increases until $C_Y = 1/\mu$ and decreases thereafter. It follows from Proposition 1 that when $Y(C_Y) = \frac{K}{\mu e} \leq q_0/(1-q_0)$, the government offers full exposure at the onset of hostility, and the adversarial firm chooses $c = \min\{X, C_Y\}$ which maximizes (10). We can see this in Proposition 2, where, for $\frac{K}{\mu e} \leq q_0/(1-q_0)$, $C_H = C_Y$, so equation (8) reduces to the degenerate distribution $H_X^* = \delta_{\min\{X,C_Y\}}$.

When $K/(\mu e) \geq q_0/(1-q_0)$, the equilibrium hostility contract screens capacity stocks sequentially over $[\underline{c}_X, \bar{c}_X]$. It begins with the lowest on-path capacity, C_Y, and gradually raises stakes so that progressively higher-capacity firms begin undermining. Consider capacity stocks c and $c+\delta$ where δ is arbitrarily small. For the firm to be indifferent, their expected hostility payoffs must be equal:

$$e^{-\mu c} u^h(\mathcal{C}_X^{h*}, c) = e^{-\mu(c+\delta)} u^h(\mathcal{C}_X^{h*}, c+\delta) \quad \iff \quad \frac{u^h(\mathcal{C}_X^{h*}, c+\delta)}{u^h(\mathcal{C}_X^{h*}, c)} = e^{\mu\delta}.$$

The contract implements this condition by making the higher-capacity firm wait longer before undermining but giving it a correspondingly larger payoff once it does. The payoffs are, therefore,

$$u^h(\mathcal{C}_X^{h*}, c) = u^h(\mathcal{C}_X^{h*}, C_Y) e^{\mu(c-C_Y)} \tag{11}$$

Thus, the higher conditional hostility payoff from capacity exactly offsets its lower probability of surviving peace-time detection which removes the incentives for deviations. These conditions produce the constant interior density, and hence the linear segment of the CDF, in (8). When $C_Y < X < C_H$, the capacity constraint truncates the support at X, so the probability mass that would otherwise be assigned to higher capacities accumulates in an atom at X. As X increases, this atom shrinks and disappears when $X = C_H$.

The government's ex ante hostility payoff, $\Psi(X)$, characterized in Proposition 2, incorporates both the effect of cumulative exposure granted before hostility and that of the subsequent equilibrium hostility contract. As Figure 1 illustrates, the government's marginal ex ante hostility payoff is negative until C_H. Consequently, its ex ante hostility payoff declines with

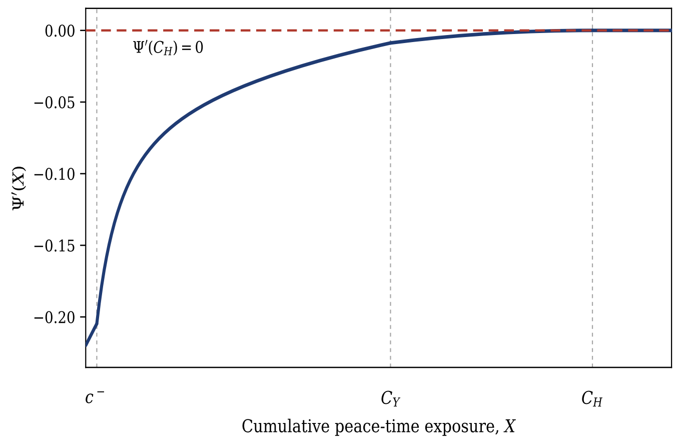

Figure 1: Government's marginal ex ante hostility payoff. The derivative $\Psi'(X)$ is negative for $X < C_H$, increases toward zero as exposure accumulates, and equals zero for $X \geq C_H$. Parameters: $r = 0.02$, $\gamma = 1.00$, $q_0 = 0.15$, $K = 0.30$, and $\mu = 0.20$.

cumulative exposure until C_H. Below c^-, additional exposure increases the capacity of an undetected adversarial firm under immediate full exposure. At c^-, the government instead begins screening: further exposure then lengthens the screening time and delays full access for the commercial firm. Once X exceeds C_Y, additional exposure no longer shifts the entire capacity distribution upward but only expands its upper tail, so the marginal decline, $-\Psi'(X)$, becomes smaller. At C_H, the firm stops building capacity, leaving both the capacity distribution and the hostility contract unchanged, $\Psi'(X) = 0$.

Proposition 2 characterizes the equilibrium separately for each cumulative-exposure level X. In the original dynamic game, however, the firm does not choose a separate distribution at every possible hostility date. A single capacity-building strategy must generate the entire family $\{H_X^*\}_{X \geq 0}$ as exposure accumulates and must remain optimal when hostility can arrive at any time. The next lemma establishes this dynamic implementation.

Lemma 2. *When geopolitical risk is persistent ($\lambda > 0$), every exposure path x^p admits a feasible equilibrium capacity process $(C_t^*)_{t \geq 0}$ such that $C_t^* \sim H_{X_t}^*$ for every t. Moreover, every PBE induces $H_t = H_{X_t}^*$ along the no-detection history, where H_X^* is characterized in Proposition 2.*

To understand the construction of the equilibrium capacity process underlying Lemma 2, recall (from Proposition 2) that the lower bound of the support of $H^*_{X_t}$ is $\underline{c}_{X_t} = \min\{X_t, C_Y\}$. Since X_t is also the largest feasible capacity, any equilibrium strategy must set $\alpha_t = 1$ until cumulative exposure reaches C_Y. Between C_Y and C_H, the firm randomizes sequentially between remaining at C_Y, continuing to build along the upper boundary $C_t = X_t$, and moving through the interior of the support. The switching rates are state dependent and assign probability across these paths so that the induced distribution remains $H^*_{X_t}$. These choices are sequentially rational because every capacity in the support yields the same survival-weighted hostility payoff under the corresponding equilibrium contract. When X_t reaches C_H, the firm stops building any capacity.

Having characterized the firm's equilibrium capacity process, we now turn to the government's choice of the exposure path.

Peace-time Exposure

If hostility begins at t, Proposition 2 and Lemma 2 imply that the government's expected hostility continuation is $\Psi(X_t)$. Since peace survives through t with probability $e^{-\lambda t}$, the government's peace-time contracting problem is therefore

$$W(x^p) = \int_0^\infty e^{-(r+\lambda)t} \left[x_t^p + \lambda\Psi(X_t)\right] dt, \qquad \dot{X}_t = x_t^p, \quad X_0 = 0, \quad x_t^p \in [\phi, 1]. \tag{12}$$

The first term is the flow payoff obtained while peace continues. The second is the hostility continuation, weighted by the instantaneous arrival rate. Because the payoff depends on the peace-time history only through X_t, the government's optimal continuation value, $V(X)$, satisfies the following Hamilton–Jacobi–Bellman (HJB) equation

$$(r+\lambda)V(X) = \max_{x\in[\phi,1]} \left\{x + \lambda\Psi(X) + xV'(X)\right\}. \tag{13}$$

The maximand is linear in current exposure, so the optimal peace-time exposure sets $x_t = 1$ when $1 + V'(X_t) > 0$, and minimal exposure, $x_t = \phi$, when $1 + V'(X_t) < 0$. Any $x_t \in [\phi, 1]$

is optimal when $1 + V'(X_t) = 0$.[10] The remaining task is to determine how the exposure choice switches between these two levels.

The switching condition depends on how cumulative peace-time exposure affects the hostility continuation. Define

$$G(X) = -\Psi'(X) \tag{14}$$

as the marginal security cost of exposure. This captures the reduction in the government's expected hostility payoff caused by an additional unit of cumulative exposure. As Figure 1 shows, this marginal cost is decreasing in X and reaches zero at C_H because, in equilibrium, the firm's capacity stock never exceeds C_H (Proposition 2). Thus, the government eventually grants full exposure.

To determine if full exposure becomes optimal before C_H, consider a cumulative exposure $X \leq C_H$ such that it grants full exposure thereafter. The corresponding continuation value is

$$V(X) = \int_0^\infty e^{-(r+\lambda)t} \left[1 + \lambda\Psi(X+t)\right] dt.$$

Because the firm's capacity stock, in equilibrium, never exceeds C_H (Proposition 2), the marginal security cost from additional exposure beyond C_H is zero, i.e., $G(s) = \Psi'(s) = 0$ for $s \geq C_H$. Differentiating $V'(X)$ and changing variables to $s = X + t$ gives

$$-V'(X) = \lambda \int_0^\infty e^{-(r+\lambda)t} G(X+t)\, dt = \lambda \int_X^{C_H} e^{-(r+\lambda)(s-X)} G(s)\, ds. \tag{15}$$

Equation (15) denotes the discounted expected security cost from an additional unit of exposure when full exposure is granted thereafter. The government compares this loss with the marginal flow benefit of increasing exposure, which is one. This comparison determines the optimal exposure path characterized in the following proposition.

Proposition 3. *Suppose constant $\lambda > 0$. The government's optimal peace-time stakes path*

[10] When the government is indifferent, we select full exposure. This tie-breaking convention does not affect equilibrium payoffs or the exposure path almost everywhere.

can have at most one switch, $\tilde{\tau}^*$ *such that*

$$x_t^{p*} = \begin{cases} \phi, & t < \tilde{\tau}^*, \\ 1, & t \geq \tilde{\tau}^*, \end{cases}$$

where $\widetilde{\tau}^* \in (0, \overline{C}/\phi)$*, if it exists, is the unique solution to*

$$1 = \lambda \int_{\phi\widetilde{\tau}^*}^{\overline{C}} e^{-(r+\lambda)(s-\phi\widetilde{\tau}^*)}(-\Psi'(s))\, ds,$$

and $\overline{C} = 1/\mu$ *if* $\frac{K}{\mu e} \leq \frac{q_0}{1-q_0}$ *and* C_H *otherwise. Furthermore, if* $\lambda \int_0^{\overline{C}} e^{-(r+\lambda)(s)}(-\Psi'(s))\, ds \leq 1$*,* $\tilde{\tau}^* = 0$*.*

Proposition 3 shows that the government may initially restrict exposure and later liberalize. Figure 2 illustrates this single-switch exposure path: at $\widetilde{\tau}^*$, the government switches from offering minimal exposure, $x_t^{p*} = \phi$, to maximum exposure, $x_t^{p*} = 1$. The single-switch structure follows from the marginal comparison in the HJB. Because the maximand in equation (13) is linear in exposure, the government chooses either ϕ or 1. Moreover, because G decreases with cumulative exposure, as illustrated in Figure 1,[11] the discounted expected marginal security cost, $-V'(X_t)$, given by equation (15), also decreases with X_t. The government therefore chooses ϕ while this cost exceeds the unit marginal benefit of exposure and switches to 1 when the cost reaches one. Since $x_t^{p*} = \phi$ throughout the initial phase, cumulative exposure at the switching date is $X_{\widetilde{\tau}^*} = \widetilde{X} = \phi\widetilde{\tau}^*$. Importantly, this liberalization happens even as the geopolitical risk, captured by the hostility arrival rate λ, remains the same.

This result highlights the interaction among three effects of exposure: the economic effect, the capacity effect, and the information effect. The economic effect is the unit marginal peacetime benefit in equation (13). The capacity effect raises the harm that an adversarial firm can impose if hostility arrives. The information effect partially offsets this harm because capacity building increases the probability of detection, $1 - e^{-\mu c}$, and changes the government's beliefs following continued non-detection, given in equation (3). The capacity and information

[11]We also prove this formally in the appendix.

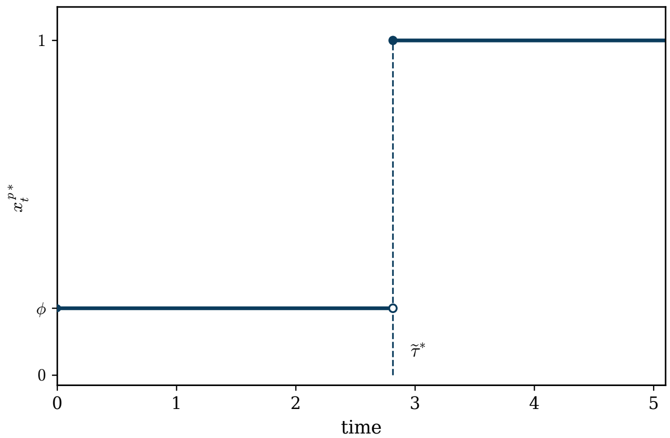

Figure 2: Peace-time stakes curve x_t when hostility arrival is random. Parameters: $r = 0.01$, $\gamma = 0.05$, $\lambda = 0.50$, $\phi = 0.20$, $q_0 = 0.70$, $K = 1.11$, and $\mu = 0.40$. The switching time is $\widetilde{\tau}^* = 2.814$.

effects jointly determine the net marginal security cost $G(X_t)$, whose discounted expected value is $-V'(X_t)$.

The capacity effect always dominates the information effect along the equilibrium path because the adversarial firm only accumulates capacity when the marginal increase in its hostility payoff exceed its loss from detection. It is easiest to see this when $K/(\mu e) \leq q_0/(1-q_0)$, where the government grants full exposure if hostility arrives. For $X_t \leq C_Y$, equation (9) gives

$$G(X_t) = -\Psi'(X_t) = \underbrace{\frac{(1-q_0)K}{r+\gamma}e^{-\mu X_t}}_{\text{capacity effect}} - \underbrace{\frac{(1-q_0)K}{r+\gamma}\mu X_t e^{-\mu X_t}}_{\text{information effect}}. \tag{16}$$

The adversarial firm only builds until C_Y (Proposition 2), where the capacity effect exactly equals the information effect. Thus, the marginal security cost is always weakly positive. However, from equation (16), the difference between the two effects decreases as the cumulative exposure increases. Thus, even though the government continues to offer minimal exposure, $x_t^{p*} = \phi$, the cumulative exposure, $X_t = \phi t$, increases with time.

This narrowing gap lowers not only the immediate marginal security cost $G(X_t)$, but also the discounted expected future security cost. Equation (15) shows that $-V'(X_t)$ aggregates the marginal security cost G over all future states at which hostility may arrive. As $X_t = \phi t$ increases during the restriction phase, these future states have lower marginal security costs. Consequently, $-V'(X_t)$ falls until it equals the unit economic benefit at $X_t = \widetilde{X}$. Beyond that point, the economic effect dominates and the government grants full exposure. Importantly, whenever $\widetilde{\tau}^* > 0$, liberalization occurs while capacity effect still dominates the information effect and the marginal security costs are still positive, $G(\widetilde{X}) > 0$: information need not eliminate the security cost; it only needs to reduce its discounted expected value sufficiently.

5 Anticipated Geopolitical Risk

The preceding analysis assumes that the risk of hostility is present from the beginning of the relationship. We now consider an anticipated increase in geopolitical risk. Hostility remains stochastic: it cannot arrive before a publicly known date T and, conditional on not yet having arrived, has a constant hazard $\lambda > 0$ thereafter.

Formally, the hazard rate is

$$\lambda(t) = \begin{cases} 0, & t < T, \\ \lambda, & t \geq T. \end{cases}$$

Thus, T marks the onset of the risk of hostility rather than the realization of hostility itself.

Let $X_T = \int_0^T x_s^p \, ds$ denote the cumulative exposure when geopolitical risk begins. Because the hostility hazard is persistent from T onward, Lemma 2 implies that, along the no-detection history, the equilibrium capacity distribution satisfies $H_t = H^*_{X_t}$ for every $t \geq T$. From T onward, the government faces the persistent-risk problem characterized in the preceding section, starting from state X_T with continuation value $V(X_T)$. Thus, for any feasible level of cumulative exposure X_T, the following lemma characterizes its optimal allocation over the pre-T period.

Lemma 3 (Front-loaded Exposure)**.** *Fix $X \in [\phi T, T]$. Among all admissible exposure paths $x^p \in \mathcal{X}$ satisfying $X_T = \int_0^T x_t^p \, dt = X$, the government's discounted peace payoff $\int_0^T e^{-rt} x_t^p \, dt$*

is maximized by the bang–bang (front-loaded) path

$$x_t^B(X) = \begin{cases} 1, & 0 \le t < \tau(X), \\ \phi, & \tau(X) \le t < T, \end{cases}$$

where $\tau(X) = \dfrac{X - \phi T}{1 - \phi} \in [0, T]$.

The logic behind the lemma follows directly from discounting. Holding X_T fixed, shifting exposure from a later time to an earlier one changes neither the equilibrium capacity distribution nor the information inherited at T, and therefore leaves the continuation value $V(X)$ unchanged. It does, however, increase the government's discounted pre-T payoff. This yields the front-loaded path in Lemma 3.

Lemma 3 reduces the government's pre-T problem to choosing cumulative exposure $X \in [\phi T, T]$. Under the front-loaded path, its payoff is

$$W_A(X, T) = \frac{1 - e^{-r\tau(X)}}{r} + \phi \frac{e^{-r\tau(X)} - e^{-rT}}{r} + e^{-rT} V(X), \tag{17}$$

where $\tau(X) = \frac{X - \phi T}{1 - \phi}$. The first term is the peace-time benefit from full exposure before $\tau(X)$, the second is the benefit from minimal exposure between $\tau(X)$ and T, and the final term is the discounted value of entering the persistent-risk period with cumulative exposure X.

To characterize the optimal exposure before T, consider the effect of an additional unit of cumulative exposure X. By Lemma 3, this additional exposure extends the initial full-exposure phase and produces a marginal pre-T benefit of $e^{-r\tau(X)}$. It also changes the discounted continuation value at T by $e^{-rT} V'(X)$. When the optimal choice is interior, these two effects balance:

$$-V'(X) = e^{r[T - \tau(X)]}. \tag{18}$$

Here, $-V'(X)$ is the expected marginal security cost of entering the risky period at T with an additional unit of exposure, while the right-hand side in equation (18) is the marginal peace-time benefit of exposure expressed in time-T units. For the remainder of the analysis, we maintain the following assumption on the minimum feasible exposure during peace.

Assumption 1 (Minimal exposure)**.** *We assume that the minimal exposure during peace, ϕ is sufficiently small. Specifically,*

$$0 < \phi \ \leq \ \min\left\{1 - \frac{q_0\gamma}{r+\gamma},\ \ 1 - \frac{r}{\lambda\big[G(\widetilde{X}) - 1\big]}\right\},$$

where $\widetilde{X} = \phi\widetilde{\tau}^*$ and $\widetilde{\tau}^*$ is defined in Proposition 3.

Substantively, this assumption implies that the minimum feasible exposure that government can restrict the foreign firms to is sufficiently small. Technically, for any fixed T, the assumption implies that the slope of $W_A(X,T)$ can cross from positive to negative at most once as X increases.[12] Hence, $W_A(X,T)$ has at most one interior local maximum, $X^I(T)$, and any global optimum must belong to the set $\{\phi T, X^I(T), T\}$.

To compare the three candidate choices as T varies, define the first payoff-crossing dates

$$\begin{aligned} T_{IL}^A &:= \inf\left\{T > 0 : W_A\big(X^I(T), T\big) - W_A(\phi T, T) \geq 0\right\}, \\ T_{FL}^A &:= \inf\left\{T > 0 : W_A(T,T) - W_A(\phi T, T) \geq 0\right\}, \\ T_{FI}^A &:= \inf\left\{T > 0 : W_A(T,T) - W_A\big(X^I(T), T\big) \geq 0\right\}, \end{aligned} \tag{19}$$

where the definitions involving $X^I(T)$ apply only when it exists. As we show in the appendix, T varies, these pairwise payoff differences are single-crossing in T, yielding the threshold characterization in the following proposition.

Proposition 4. *Suppose that the hostility hazard is zero before T and equals a constant $\lambda > 0$ thereafter. Define $X_T(\tau) = \phi T + (1-\phi)\tau$, and $\widetilde{X} = \phi\widetilde{\tau}^*$, where $\widetilde{\tau}^*$ is defined in Proposition 3. Then, under Assumption 1, the government's optimal peace-time exposure*

[12] Intuitively, because $\tau'(X) = 1/(1-\phi)$, a small ϕ makes the marginal benefit from extending the initial full-exposure phase decline relatively slowly with X. This ensures that the marginal peace-time benefit $e^{-r\tau(X)}$ can cross $e^{-rT}[-V'(X)]$ from above at most once.

path is

$$x_t^{p*} = \begin{cases} 1, & t < \tau_1^*, \\ \phi, & \tau_1^* \leq t < \tau_2^*, \\ 1, & t \geq \tau_2^*, \end{cases}.$$

where,

- *the two switching times* $0 \leq \tau_1^* \leq T \leq \tau_2^*$ *are*

$$\tau_1^* = \begin{cases} 0, & T < \underline{T}^A, \\ \tau^I(T), & \underline{T}^A \leq T < \overline{T}^A, \\ T, & T \geq \overline{T}^A, \end{cases}$$

  and $\tau_2^* = T + \frac{1}{\phi} \max\left\{0, \widetilde{X} - X_T(\tau_1^*)\right\}$, *respectively.*

- *When it exists,* $\tau^I(T) \in (0, T)$ *is unique and satisfies*

$$e^{-\frac{r+\lambda}{\phi}\left[\widetilde{X} - X_T(\tau^I(T))\right]} + \frac{\lambda}{\phi} \int_{X_T(\tau^I(T))}^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}\left[s - X_T(\tau^I(T))\right]} G(s)\, ds = e^{r\left(T - \tau^I(T)\right)},$$

  where $G(\cdot)$ *is defined in equation (14). The interior exposure is* $X^I(T) = X_T(\tau^I(T))$.

- *If* $\tau^I(T)$ *exists, the thresholds* $0 \leq \underline{T}^A \leq \overline{T}^A$ *are* $\underline{T}^A = \min\{T_{IL}^A,\ T_{FL}^A\}$, *and* $\overline{T}^A = \max\{T_{FL}^A, T_{FI}^A\}$, *where* T_{IL}^A, T_{FL}^A, *and* T_{FI}^A *are defined in equation (19). If* $\tau^I(T)$ *does not exists, then* $\underline{T}^A = \overline{T}^A = T_{FL}^A$.

Setting $T = 0$ in Proposition 4 recovers the persistent risk result characterized in Proposition 3. Because $0 \leq \tau_1^* \leq T$, setting $T = 0$ implies $\tau_1^* = 0$ and $X_T(\tau_1^*) = \phi T + (1-\phi)\tau_1^* = 0$. The second switching time then becomes $\tau_2^* = \widetilde{X}/\phi = \widetilde{\tau}^*$. Hence, the government grants minimal exposure until $\widetilde{\tau}^*$ and full exposure thereafter, exactly as characterized in Proposition 3.

The two switches in Proposition 4 arise when $T > 0$ and the cumulative exposure that the government chooses to enter the risky period with, along the no detection history, is $X^I(T) \in (\phi T, T)$. By Lemma 3, the government implements this choice by granting full exposure first and minimal exposure for the remainder of the pre-T period. The first switch is, therefore, $\tau_1^* = \left(X^I(T) - \phi T\right)/(1-\phi) \in (0, T)$, so that $X^I(T) = \phi T + (1-\phi)\tau_1^* = X_T(\tau_1^*)$. At this interior optimum, equation (18) implies $-V'(X_T(\tau_1^*)) = e^{r[T-\tau_1^*]} > 1$ because $\tau_1^* < T$. In the period with geopolitical risks ($\lambda > 0$), it follows from Proposition 3 that full exposure is optimal when the cumulative exposure exceeds $X \geq \widetilde{X} = \phi\widetilde{\tau}^*$, when the expected marginal security costs equal the unit economic benefit, $-V'(\widetilde{X}) = 1$. Conditional on peace continuing, the government, therefore, continues granting minimal exposure after T until cumulative exposure reaches $\widetilde{X}$. This occurs at $\tau_2^* = T + [\widetilde{X} - X^I(T)]/\phi$, when the government returns to full exposure. Figure 3 illustrates the resulting integrate-retrench-liberalize ($1 \to \phi \to 1$) exposure path.

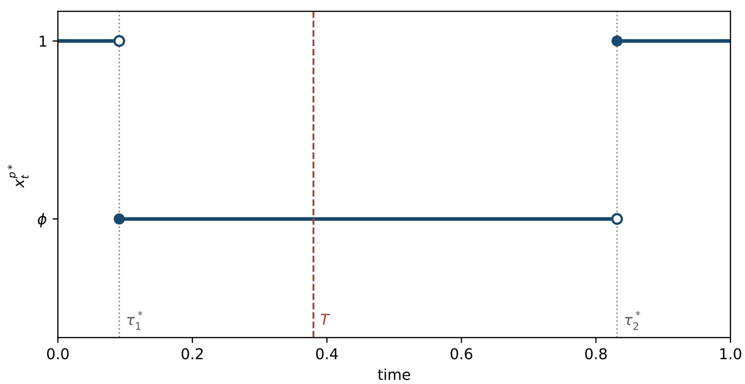

Figure 3: Peace-time stakes curve x_t under anticipated geopolitical risk. The hostility hazard is zero before $T = 0.38$ and equals $\lambda = 5.00$ thereafter. Parameters: $r = 0.10$, $\gamma = 0.01$, $\phi = 0.40$, $q_0 = 0.90$, $K = 1.19$, and $\mu = 0.05$. The government switches from full to minimal stakes at $\tau_1^* = 0.091$ and returns to full stakes at $\tau_2^* = 0.831$.

Thus, Proposition 4 reveals an interesting reversal relative to the persistent-risk case in Proposition 3. Under persistent risk, the government initially restricts exposure and subsequently liberalizes; when risk is anticipated, it may instead grant full exposure, retrench, and then liberalize again. For a fixed X_T, front-loading changes neither the capacity accumulated nor the information acquired until T, and therefore, the expected marginal security cost at T, $-V'(X_T)$ remains unchanged. Instead, it raises the economic effect by delivering the economic gains from raising exposure earlier while hostility cannot occur.

The initial full-exposure phase nevertheless moves cumulative exposure toward $\widetilde{X}$ at the maximum rate, shortening the subsequent retrenchment phase.

Corollary 1. *Retrenchment is weakly shorter under anticipated geopolitical risk than under persistent geopolitical risk.*

Under persistent risk, the government offers the cumulative exposure upto $\widetilde{X}$ at the restricted rate ϕ. Under anticipated risk, it reaches $\tau_1^* > 0$ at the maximum rate before retrenchment begins, leaving only the interval from τ_1^* to $\widetilde{X}$ to be offered at the rate ϕ. Anticipation therefore delays the onset of restriction relative to persistent risk while also bringing liberalization forward, making the total retrenchment period shorter.

6 Predictable Hostility

We now analyze a setting where hostility arrives at some publicly known time T. This setup represents cases where countries can anticipate the arrival of hostility. Predictable hostility is the limiting case of anticipated geopolitical risk in which the post-T hostility hazard becomes arbitrarily large: as $\lambda \to \infty$, hostility begins at T with certainty. Thus, $T_h = T$.

From Lemma 2, the strategic firm's peace-time behavior matters for the government only through the induced distribution H_X^*. Thus, for an exposure path x^p with cumulative exposure X, the government's expected payoff is

$$W(x^p, H_X^*) = \int_0^{T_h} e^{-rt} x_t^p \, dt + e^{-rT_h} \Psi(X). \tag{20}$$

Compared to the anticipated-risk problem in equation (17), $V(X)$ is replaced by $\Psi(X)$, because hostility begins immediately at $T = T_h$. By Lemma 3, the government, therefore, implements any given $X \in [\phi T_h, T_h]$ by offering full exposure until $\tau(X)$ and minimal exposure thereafter. Substituting this path into equation (20), the government chooses cumulative exposure X to maximize

$$W(X, T_h) = \frac{1 - e^{-r\tau(X)}}{r} + \phi \frac{e^{-r\tau(X)} - e^{-rT_h}}{r} + e^{-rT_h} \Psi(X). \tag{21}$$

The following proposition characterizes the equilibrium exposure:

Proposition 5. *Suppose hostility arrives at T_h that is known. Define $X_{T_h}(\tau) = \phi T_h + (1 - \phi)\tau$. Under Assumption 1, the government's equilibrium exposure path during peace is*

$$x_t^{p*} = \begin{cases} 1, & t < \tau^*, \\ \phi, & \tau^* \le t < T_h, \end{cases}$$

where,

$$\tau^* = \begin{cases} 0, & T_h < \underline{T}', \\ \tau^I(T_h), & \underline{T}' \le T_h < \overline{T}', \\ T_h, & T_h \ge \overline{T}'. \end{cases}$$

- *When $\tau^I(T_h) \in (0, T_h)$ exists, it is unique and satisfies*

  $$\frac{(1-q_0)K}{r+\gamma} e^{-\mu X_{T_h}(\tau^I(T_h))}(1 - \mu X_{T_h}(\tau^I(T_h))) = e^{r[T_h - \tau^I(T_h)]},$$

  where $X^I(T_h) = X_{T_h}(\tau^I(T_h))$.

- *If $\tau^I(T_h)$ exists, the thresholds are $\underline{T}' = \min\{T'_{IL}, T'_{FL}\}$ and $\overline{T}' = \max\{T'_{FL}, T'_{FI}\}$ where T'_{IL}, T'_{FL}, and T'_{FI} uniquely solve,*

  $W(\phi T_h, T_h) = W(X_{T_h}(\tau^I(T_h)), T_h); W(T_h, T_h) = W(\phi T_h, T_h); W(T_h, T_h) = W(X_{T_h}(\tau^I(T_h)), T_h),$

  respectively. If $\tau^I(T_h)$ does not exist, the thresholds are $\underline{T}' = \overline{T}' = T'_{FL}$.

The structure of Proposition 5 matches closely with that of Proposition 4. In fact, it is the limiting case of the anticipated geopolitical risk. First, note that as $\lambda \to \infty$, the second cutoff τ_2^* disappears. That is because $\tau_2^* \ge T$. In the predictable hostility case, hostility arrives almost surely at $T = T_h$. Thus, hostility arrives before the government could potentially liberalize during peace.

The first cutoff remains but its condition simplifies. In Proposition 4, the interior cutoff

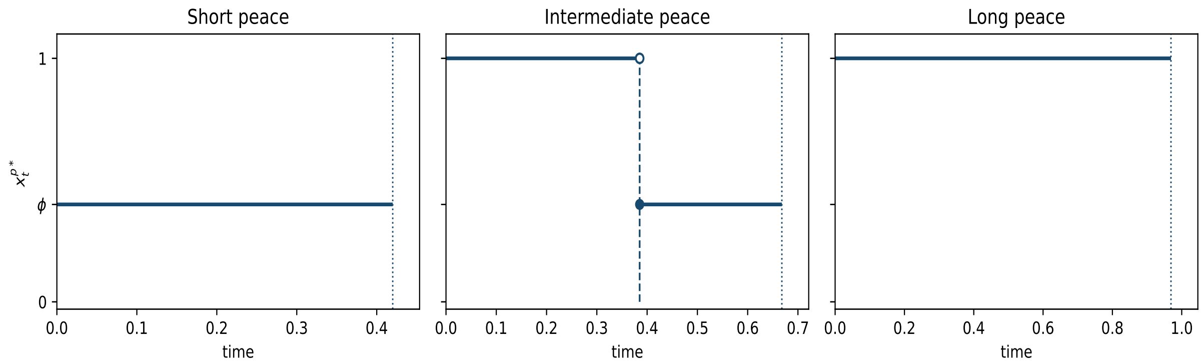

Figure 4: Peace-time stakes curve x_t^{p*} under different under short, intermediate, and long periods of predictable peace. Parameters: $r = 0.10$, $\gamma = 0.01$, $\phi = 0.40$, $q_0 = 0.90$, $K = 1.19$, and $\mu = 0.05$. The three panels correspond to $T_h = 0.420$ (short peace), $T_h = 0.668$ (intermediate peace), and $T_h = 0.969$ (long peace).

satisfies

$$e^{-\frac{r+\lambda}{\phi}\left[\widetilde{X}-X_T(\tau^I(T))\right]} + \frac{\lambda}{\phi}\int_{X_T(\tau^I(T))}^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}\left[s-X_T(\tau^I(T))\right]}G(s)\,ds = e^{r\left(T-\tau^I(T)\right)}.$$

As $\lambda \to \infty$, the first term of the LHS vanishes and the second term converges to

$$\lim_{\lambda\to\infty}\left[\frac{\lambda}{\phi}\int_{X_T(\tau^I(T))}^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}(s-X_T(\tau^I(T)))}G(s)\,ds\right] = G(X_{T_h}(\tau^I(T_h))).$$

The second equality follows because the expression for G follows the first branch in equation (14) which yields the condition in Proposition 5. The predictable-hostility policy therefore restricts exposure throughout when T_h is short, grants full exposure before retrenching when T_h is intermediate, and maintains full exposure throughout when T_h is sufficiently long, as illustrated in Figure 4.

7 Conclusion

This paper studies how a government should regulate a foreign firm's access over time when that access is economically valuable during peace but may facilitate harm under hostility. Exposure produces three effects. It generates an immediate economic benefit, gives an adversarial firm an opportunity to build capacity, and produces information through the possibility that this activity is detected. Because capacity and information arise from the same activity, the government cannot obtain information without enabling the building of capacity whose security costs it seeks to limit. Investment screening is therefore a dynamic problem of managing exposure rather than simply deciding whether to permit or prohibit foreign investment.

The analysis in this paper reveals some interesting insights. When geopolitical risk is persistent, the government may initially restrict exposure and subsequently liberalize (Proposition 3). This liberalization occurs even though geopolitical risk remains unchanged and the firm's type remains uncertain. Because the adversarial firm builds capacity only while the marginal increase in its hostility payoff exceeds the expected loss due to detection, which leads to its exclusion from the economy, the capacity effect weakly dominates the information effect along the equilibrium path. Information nevertheless narrows the difference between them as exposure accumulates. Thus, the expected marginal security cost subsequently declines until the economic benefit of exposure outweighs it. Liberalization, therefore, occurs even as the information effect never dominates the capacity effect and the government remains uncertain about the firm's type.

The ordering reverses when geopolitical risk begins only after an anticipated period of peace. For any cumulative exposure carried into the risky period, front-loading changes neither the capacity distribution nor the information inherited when risk begins. However, it allows the government to collect the economic benefits of exposure sooner. Therefore, the government may grant full exposure initially, retrench before geopolitical risk begins, and liberalize again if peace continues (Proposition 4). Moving cumulative exposure toward the liberalization threshold at the maximum rate during this initial period of full exposure makes the retrenchment period weakly shorter than under persistent risk (Corollary 1).

These results shift attention from the binary decision to approve or prohibit an investment to the management of access within an active relationship. Retrenchment need not indicate that the government has detected hostile activity, while liberalization need not imply that geopolitical risk has declined or that the firm has been revealed to be commercial. The same economic interests, security concerns, and uncertainty about the firm can produce different policy sequences depending on when geopolitical risk begins.

Two extensions provide directions for future work:

- Verifiable Peace-time Signal: Peace-time detection could generate verifiable evidence which would allow the government to condition its peace-time contract on this detection. This could make continuation access or termination contingent on peace-time detection.

- Hostility Conditional on Actions: The arrival rate of hostility could depend on the capacity accumulated by the firm. Exposure would then affect not only the harm conditional on hostility but also the probability that hostility occurs. Greater capacity might encourage preventive action or embolden an adversary, but it might also deter hostility, so this relationship need not be monotonic. Moreover, continued peace would itself become informative about hidden capacity.

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Online Appendix

A Screening Contract Under Hostility

The following proposition describes the contract during hostility as well as the continuation payoffs

Proposition A.6 (From Theorem 1, Kolb and Madsen (2023))**.** *Suppose hostility begins at T_h, no capacity building detected during peace, the adversarial firm's capacity stock is c, and the government's beliefs over capacity is $P^C = \delta_c$. Let $t \geq 0$ denote the time elapsed since the onset of hostility. Given posterior q and effective harm $k(c)$, the optimal hostility contract, $\mathcal{C}^{h*}(q, k(c))$, is a stringent contract (termination upon detection of undermining) that is pinned down by q and $k(c)$ such that,*

(i) High Posteriors: *If $q \geq \frac{k(c)-1}{k(c)}$, the government sets $x_t = 1$ for all $t \geq 0$ and the adversarial firm always undermines with full intensity, $\beta_t = 1$ for all $t \geq 0$. The adversarial firm and government's hostility continuation payoffs are, respectively,*

$$u^h(\mathcal{C}^{h*}(q, k(c)), c) = \frac{k(c)-1}{r+\gamma} \quad ; \quad V^h(\mathcal{C}^{h*}(q, k(c)), c) = \frac{q}{r} - (1-q)\left(\frac{k(c)-1}{r+\gamma}\right)$$

(ii) Low Posteriors: *If $q \leq \frac{k(c)-1}{k(c)}$, then, the unique optimal stakes path begins immediately in an escalation phase with the stakes growing continuously at rate $r + \frac{\gamma}{k(c)}$ followed by a discrete jump to $x_t = 1$ at time $\bar{t}_l = \frac{k(c)}{\gamma} \log\left(\frac{k(c)-1}{k(c)q}\right)$. The adversarial firm is indifferent between undermining with full intensity, $\beta_t = 1$, and not undermining at all, $\beta_t = 0$ for all $t \geq 0$. The adversarial firm and government's hostility continuation payoffs are, respectively,*

$$u^h(\mathcal{C}^{h*}(q, k(c)), c) = \frac{k(c)-1}{r+\gamma}\left(\frac{qk(c)}{k(c)-1}\right)^{1+\frac{rk(c)}{\gamma}} \quad ; \quad V^h(\mathcal{C}^{h*}(q, k(c)), c) = q\frac{\gamma}{r(r+\gamma)}\left(\frac{qk(c)}{k(c)-1}\right)^{\frac{rk(c)}{\gamma}}$$

B Proofs[13]

Proof of Lemma 1. Excluding the firm before hostility is suboptimal. Suppose the government excludes the firm at some finite T_R^p while peace continues. Becasue capacity produces no harm during peace, exclusion provides no security beenfits, instead, it causes the government to forego the peace-time economic gains, that are strictly positive, $x_t^p > \phi > 0$. If hostility arrives, the government can still exclude the firm at T_h. Thus, every finite peace-time exclusion is strictly sub-optimal. Therefore, $T_R^{p*} = \infty$. □

Proof of Proposition 1. Conditional on no peace-time detection, when the government believes that the capacity stock c, its posterior is

$$q(c) = \frac{q_0}{q_0 + (1-q_0)e^{-\mu c}}.$$

Given $k(c) = 1 + Kc$, from Proposition A.6, the government grants immediate full exposure if and only if

$$q(c) \geq \frac{k(c)-1}{k(c)} = \frac{Kc}{1+Kc},$$

which is equivalent to

$$\begin{aligned} q_0(1+Kc) &\geq Kc\left[q_0 + (1-q_0)e^{-\mu c}\right] \\ \frac{q_0}{1-q_0} &\geq Kce^{-\mu c}. \end{aligned}$$

Recall that $Y(c) = Kce^{-\mu c}$. Its derivative is $Y'(c) = Ke^{-\mu c}(1-\mu c)$. Hence, Y is strictly increasing on $[0, 1/\mu)$ and strictly decreasing on $(1/\mu, \infty)$. Thus, its unique maximizer is $C_Y = 1/\mu$ with $Y(C_Y) = \frac{K}{\mu e}$.

Consider two cases:

- If $\frac{K}{\mu e} \leq \frac{q_0}{1-q_0}$, then $Y(c) \leq q_0/(1-q_0)$ for every $c \geq 0$, so the government grants immediate full exposure for every c.
- If $\frac{K}{\mu e} \geq \frac{q_0}{1-q_0}$, because $Y(0) = 0 < \frac{q_0}{1-q_0} \leq Y(C_Y)$, continuity and strict monotonicity imply that there is a unique $c^- \in (0, C_Y]$ satisfying $Y(c^-) = \frac{q_0}{1-q_0}$. Similarly, since Y is strictly decreasing above C_Y and converges to zero, there is a unique $c^+ \in [C_Y, \infty)$

[13] I am currently revising the proofs and would appreciate any comments.

satisfying $Y(c^+) = \frac{q_0}{1-q_0}$. Thus,

$$Y(c) \leq \frac{q_0}{1-q_0} \iff c \leq c^- \text{ or } c \geq c^+.$$

When $\frac{K}{\mu e} = \frac{q_0}{1-q_0}$, $Y(C_Y) = \frac{q_0}{1-q_0}$ and single-peakedness of $Y(c)$ implies that $Y(c) < Y(C_Y) = \frac{q_0}{1-q_0}$ for all $c \neq C_Y$. Thus, $c^- = C_Y = c^+$.

Therefore, the government therefore grants immediate full exposure for $c \leq c^-$ and $c \geq c^+$, and offers the screening contract for $c \in [c^-, c^+]$.

If capacity building is detected during peace, the government learns that the firm is adversarial and therefore excludes it when hostility begins.

□

Proof of Proposition 2. We prove this in two parts. First consider the following Lemma:

Lemma B.4 (Equilibrium capacity distribution)**.** *Suppose the government provides cumulative peace-time exposure X. The equilibrium capacity distribution is*

$$H_X^*(c) = \begin{cases} 0, & c < \min\{X, C_Y\}, \\ \dfrac{1 + K\big(c - \min\{X, C_Y\}\big)}{M\big(\min\{X, C_H\}\big)}, & \min\{X, C_Y\} \leq c < \min\{X, C_H\}, \\ 1, & c \geq \min\{X, C_H\}, \end{cases}$$

where $M(z) = (1 + Kz)\big(1 - q(z)\big)$ and $C_H = \min\{c \geq C_Y : \ \mu q(c)(c + \frac{1}{K}) \geq 1\}$.

Proof. Fix a total exposure budget X. Define

$$w(c) = \mu q(c) \left(c + \frac{1}{K}\right) \tag{22}$$

Because $q(c)$ is strictly increasing, $w(c)$ is strictly increasing. Moreover,

$$w(C_Y) \geq 1 \quad \iff \quad \frac{K}{\mu e} \leq \frac{q_0}{1-q_0}.$$

Define

$$C_H = \min\{c \geq C_Y : w(c) \geq 1\}. \tag{23}$$

We have $C_H = C_Y$ in the no-screening case. In the strict screening case, $C_H > C_Y$ and is the unique solution to $w(C_H) = 1$. Then consider the following cases.

First, suppose $Y(C_Y) \leq q_0/(1-q_0)$. For any $C \leq X$, the strategic type's continuation payoff is proportional to

$$e^{-\mu C}\big(k(C) - 1\big) = Y(C).$$

Because Y is strictly increasing on $[0, C_Y]$ and strictly decreasing on (C_Y, ∞), the unique best response on $[0, X]$ is $C^* = \min\{X, C_Y\}$. Thus, the unique induced distribution is $H_X^* = \delta_{\min\{X, C_Y\}}$.

Next, suppose $Y(C_Y) > q_0/(1 - q_0)$. First suppose $X \leq C_Y$. For any $C \leq X$, the strategic type's continuation payoff is proportional to

$$e^{-\mu C}\big(k(C) - 1\big) = Y(C).$$

Because Y is strictly increasing on $[0, C_Y]$, the unique best response on $[0, X]$ is $C^* = X$. Thus, the unique induced distribution is $H_X^* = \delta_X$.

Next, suppose $X \geq C_Y$. This part is similar to the problem we study in Zalke (2026). The calculations for the optimal hostility contract as well as the distribution over C follow the same way as in the screening problem studied in Zalke (2026). Thus, the proof here shows only the calculations that are relevant to the proof here.

The best response cannot choose a total capacity less than C_Y because her continuation payoff, $e^{-\mu C}\big(k(C) - 1\big) = Y(C)$, is increasing until C_Y. So, the support of strategic type's best response must have a lower bound at C_Y.

Consider a distribution with support $[C_Y, z]$ where $z \leq X$. Let the distribution be of the form

$$dH(C) = m_Y \delta_{C_Y}(dC) + h(C)\mathbf{1}\{C \in (C_Y, z)\}\, dC + m_z \delta_z(dC),$$

where $h(C)$ denotes the density on (C_Y, z). For $C \in (C_Y, z)$, define the posterior survivor mass as the probability, conditional on no detection, that the firm is either commercial or a strategic type with total capacity strictly greater than C as

$$s_H(C) = \frac{q_0 + (1 - q_0)\int_C^z e^{-\mu y}\, dH(y)}{\Delta_H}, \tag{24}$$

where $\Delta_H = Pr(\text{no detect}|H) = q_0 + (1-q_0)\int e^{-\mu C}\, dH(C)$.

The strategic firm must get the same expected payoff over the support in its best response. Thus, for $C \in (C_Y, z)$, it must be that $e^{-\mu C} u^h(\mathcal{C}^h(H), C)$ is constant. The fixed-posterior screening equations derived in Zalke (2026) imply that for all $C \in (C_Y, z)$, this indifference condition can be written in terms of the posterior survivor mass as

$$s_H(C) = s_H(C_Y^+) e^{-\mu(C-C_Y)}. \tag{25}$$

At the lower endpoint, the government's optimal hostility contract must satisfy $k(C_Y)(1 - s_H(C_Y^+)) = 1$. This is explained in detail in the proof of Theorem 2 in Zalke (2026). Thus,

$$s_H(C_Y^+) = \frac{k(C_Y) - 1}{k(C_Y)} = \frac{K/\mu}{1 + K/\mu} = \frac{K}{K + \mu}. \tag{26}$$

The second equality follows because $k(C) = 1 + KC$ and $C_Y = 1/\mu$. Combining this with (24), the atom at the lower bound, C_Y, of the support of the posterior distribution is

$$1 - s_H(C_Y^+) = 1 - \frac{q_0 + (1-q_0)\int_{C_Y^+}^{z} e^{-\mu y}\, dH(y)}{\Delta_H} = \frac{(1-q_0) m_Y e^{-\mu C_Y}}{\Delta_H} = \frac{\mu}{K+\mu}. \tag{27}$$

Next, for $C \in (C_Y, z)$, from (24),

$$s'_H(C) = -\frac{(1-q_0) h(C) e^{-\mu C}}{\Delta_H} = -\mu s_H(C),$$

where the second equality follows from differentiating equation (25). Using (25) and $s_H(C_Y^+) = K/(K+\mu)$,

$$\begin{aligned} h(C) &= \frac{\mu \Delta_H}{1-q_0} e^{\mu C} s_H(C) = \frac{\mu \Delta_H}{1-q_0} e^{\mu C} \frac{K}{K+\mu} e^{-\mu(C-C_Y)} \\ &= \frac{\mu K}{K+\mu} \frac{\Delta_H e^{\mu C_Y}}{1-q_0} = K m_Y. \end{aligned} \tag{28}$$

The last equality follows because equation (27) implies

$$m_Y = \frac{\mu}{\mu + K} \frac{\Delta_H e^{\mu C_Y}}{1 - q_0}.$$

Therefore, for $C \in (C_Y, z)$,

$$h(C) = Km_Y. \tag{29}$$

For the atom at the upper bound of the support, z, suppose $m_z = m_Y l$. Because the total probability must be equal to one,

$$m_Y + Km_Y(z - C_Y) + m_Y l = 1,$$

which implies that

$$m_Y = \frac{1}{1 + K(z - C_Y) + l}. \tag{30}$$

From (24), the posterior survivor mass just below z is

$$s_H(z^-) = \frac{q_0 + (1-q_0)m_z e^{-\mu z}}{\Delta_H} = \frac{K}{K+\mu} e^{-\mu(z - C_Y)},$$

where the second equality follows from equation (25). It follows from equation (27) that

$$\Delta_H = (1-q_0)m_Y e^{-\mu C_Y} \frac{K+\mu}{\mu}.$$

Equating the two expressions for $s_H(z^-)$ and substituting Δ_H,

$$\begin{aligned}
q_0 + (1-q_0)m_Y l e^{-\mu z} &= (1-q_0)m_Y \frac{K}{\mu} e^{-\mu z} \\
q_0 &= (1-q_0)m_Y e^{-\mu z}\left(\frac{K}{\mu} - l\right) \\
q_0 &= (1-q_0)\frac{1}{1 + K(z - C_Y) + l} e^{-\mu z}\left(\frac{K}{\mu} - l\right) \qquad \text{(from equation (30))} \\
l\big(q_0 + (1-q_0)e^{-\mu z}\big) &= (1-q_0)\frac{K}{\mu}e^{-\mu z} - q_0\big(1 + K(z - C_Y)\big).
\end{aligned}$$

Therefore,

$$\begin{aligned}
l(z) &= \frac{(1-q_0)\frac{K}{\mu}e^{-\mu z} - q_0\big(1 + K(z - C_Y)\big)}{q_0 + (1-q_0)e^{-\mu z}} \\
&= \frac{K}{\mu}\left[1 - \mu q(z)\left(z + \frac{1}{K}\right)\right].
\end{aligned} \tag{31}$$

Consider the numerator in the first line of (31). Denote it by $N(z)$. At $z = C_Y$,

$$N(C_Y) = (1 - q_0)\frac{K}{\mu}e^{-\mu C_Y} - q_0.$$

Because $Y(C_Y) = e^{-\mu C_Y}K/\mu$ and $Y(C_Y) > q_0/(1-q_0)$, $N(C_Y) > 0$. Moreover,

$$N'(z) = -(1 - q_0)Ke^{-\mu z} - q_0 K < 0.$$

So N is strictly decreasing. Thus, there exists a unique $C_H > C_Y$ such that $l(C_H) = 0$. From (31), this is equivalent to

$$q(C_H)\left(C_H + \frac{1}{K}\right) = \frac{1}{\mu}.$$

When $X \geq C_H$, the optimal hostility contract offered by the government and the strategic firm's choice over C are given by the expressions in Theorem 2 in Zalke (2026). Thus, the strategic firm's equilibrium distribution is of the form

$$dH_X^*(C) = \frac{1}{1 + K(C_H - C_Y)}\delta_{C_Y}(dC) + \frac{K}{1 + K(C_H - C_Y)}\mathbf{1}\{C \in (C_Y, C_H]\}\, dC. \tag{32}$$

We can denote the stakes in terms of what capacity is screened at time t. Let κ_t denote the total capacity that is indifferent (or screened) at time t. That is, the agent who built capacity κ_t is indifferent between undermining and not at time t. Thus, the optimal stakes curve during hostility follows:

- The cutoff starts at screening capacity C_Y, and the cutoff path satisfies
$$\kappa_t = \frac{1}{K}\left(\left[(1 + KC_Y)^2 + \frac{2\gamma K}{\mu}t\right]^{1/2} - 1\right).$$

- The stakes curve is
$$x_t^h = \begin{cases} \dfrac{\gamma K \kappa_t}{(r+\gamma)(1 + K\kappa_t)}e^{-\mu(C_H - \kappa_t)\left[1 + \frac{r}{2\gamma}(2 + K(C_H + \kappa_t))\right]}, & t < \dfrac{\mu}{2\gamma K}\left[(1 + KC_H)^2 - (1 + KC_Y)^2\right], \\ 1, & t \geq \dfrac{\mu}{2\gamma K}\left[(1 + KC_H)^2 - (1 + KC_Y)^2\right]. \end{cases}$$

When $X < C_H$, the indifference equations remain the same along the C_H envelope at the end before it jumps. In this case, $z = X$ and denote

$$M(X) = 1 + K(X - C_Y) + l(X) = (1 + KX)(1 - q(X)).$$

From the calculations above, the distribution over total capacity must be of the form

$$dH_X^*(C) = \frac{1}{M(X)}\delta_{C_Y}(dC) + \frac{K}{M(X)}\mathbf{1}\{C \in (C_Y, X)\}\, dC + \frac{l(X)}{M(X)}\delta_X(dC). \tag{33}$$

The hostility contract associated with this H_X^* is the truncated version of the hidden-harm moving-cutoff contract. That is, the stakes are similar as above so that screening starts at C_Y and evolves according to the same expression so that

$$\kappa_t = \frac{1}{K}\left(\left[(1 + KC_Y)^2 + \frac{2\gamma K}{\mu}t\right]^{1/2} - 1\right), \tag{34}$$

with the cutoff reaching $z = X$ at

$$T(X) = \frac{\mu}{2\gamma K}\left[(1 + KX)^2 - (1 + KC_Y)^2\right]. \tag{35}$$

However, because of the atom at $z = X$, the optimal stakes continue along the X-indifference envelope. This follows from the results in Kolb and Madsen (2023). The stakes continue along this path until it jumps to one and the duration of this follows from the indifference of the continuation value just before the stakes jump to 1. Thus, the duration of this is

$$T_A(X) = \frac{1 + KX}{\gamma}\log\left(\frac{q_0 + (1 - q_0)e^{-\mu X}}{q_0\mu\left(X + \frac{1}{K}\right)}\right). \tag{36}$$

The associated hostility stakes curve is therefore

$$x_t^h(X) = \begin{cases} \dfrac{\gamma K\kappa_t}{(r+\gamma)(1+K\kappa_t)} e^{\left\{-\mu(X-\kappa_t) - \frac{r\mu}{2\gamma K}\left[(1+KX)^2 - (1+K\kappa_t)^2\right] - \left[r + \frac{\gamma}{1+KX}\right]T_A(X)\right\}}, & 0 \le t < T(X), \\ \dfrac{\gamma KX}{(r+\gamma)(1+KX)} e^{-\left(r + \frac{\gamma}{1+KX}\right)(T_A(X) - (t - T(X)))}, & t \in [T(X), T(X) + T_A(X)) \\ 1, & t \ge T(X) + T_A(X). \end{cases} \tag{37}$$

Note that the strategic firm is indifferent over this support and she has no profitable deviation outside of the support $[C_Y, \min\{X, C_H\}]$. The government's hostility contract is the optimal contract it can offer given the distribution over capacity. Thus, this is an equilibrium.

The proof for uniqueness is the same as the proof for uniqueness in Theorem 2 in Zalke (2026). First note that we showed above that the lower bound of the support of the distribution over C must be above C_Y whenever $X \geq C_Y$. Thus, it follows from the calculations above that the strategic type will never choose a capacity below C_Y. When $C_Y < X < C_H$, feasibility rules out choosing a capacity greater than X. For $X > C_H$, the strategic firm can feasibly choose a $C > C_H$; however, any such deviation gives the strategic firm a strictly lower payoff.

Finally, using the same approach as in the proof of Theorem 2 in Zalke (2026), there cannot be any equilibria where the distribution over C has an interior atom or a gap in the support. Suppose, by contradiction, there exists such an equilibrium with an interior atom. Suppose H has an atom of size $m > 0$ at some $C_0 \in (C_Y, z)$. In such an equilibrium, the hostility contract that the government offers must wait on the C_0 indifference envelope. However, in that case, the strategic type who builds a capacity of C_0 can do better by instead building a capacity of $C_0 + \epsilon$. Thus, this cannot be an equilibrium.

To rule out a gap in the support, suppose there is a gap (a, b) in the equilibrium distribution. It follows that the strategic type must get a strictly lower payoff for choosing a capacity $C \in (a, b)$. Indifference implies that $e^{-\mu a} U^N(a) = e^{-\mu b} U^N(b) \implies U^N(b) = e^{\mu(b-a)} U^N(a)$. Also, $U^N(C)$ is affine over the gap. However, the RHS is strictly convex. So there must exist some $C \in (a, b)$ such that $e^{-\mu C} U^N(C) > e^{-\mu a} U^N(a)$, which implies that the strategic firm has a profitable deviation. Thus, this cannot be an equilibrium.

Stating the distribution as a CDF, we get

$$H_X^*(c) = \begin{cases} 0, & c < \min\{X, C_Y\}, \\ \dfrac{1 + K\big(c - \min\{X, C_Y\}\big)}{M\big(\min\{X, C_H\}\big)}, & \min\{X, C_Y\} \leq c < \min\{X, C_H\}, \\ 1, & c \geq \min\{X, C_H\}, \end{cases}$$

where $M(z) = (1 + Kz)\big(1 - q(z)\big)$ and $C_H = \min\{c \geq C_Y :\ \mu q(c)(c + \frac{1}{K}) \geq 1\}$.

This completes the proof.

□

Next, consider the following lemma:

Lemma B.5. *Write $\kappa = \min\{X, C_Y\}$ and $z = \min\{X, C_H\}$, and let*

$$D(X) = \frac{\mu}{2\gamma K}\Big[(1+Kz)^2 - (1+K\kappa)^2\Big] - \frac{1+Kz}{\gamma}\log\frac{w(z)}{\mu\kappa}, \qquad w(c) = \mu q(c)\Big(c + \frac{1}{K}\Big).$$

The government's expected hostility payoff as a function of total exposure X is

$$\Psi(X) = \begin{cases} \dfrac{q_0}{r} - \dfrac{1-q_0}{r+\gamma}\,K\kappa e^{-\mu\kappa}, & X \le c^-, \\ \dfrac{q_0\gamma}{r(r+\gamma)}\,e^{-rD(X)}, & X \ge c^-, \end{cases}$$

where c^- is as characterized in Proposition 1, with $c^- = \infty$ when $K/\mu e \le q_0/(1-q_0)$.

Proof. Fix a total exposure X. Define $w(c)$ and C_H as we do in equations (22) and (23), respectively.

First, suppose $\frac{K}{\mu e} \le \frac{q_0}{1-q_0}$. By Lemma B.4, the adversarial firm's equilibrium capacity distribution is degenerate at $c_X = \min\left\{X, \frac{1}{\mu}\right\}$. Moreover, from Proposition 1, the government grants full exposure during hostility for every inherited capacity. Hence,

$$\begin{aligned}\Psi(X) &= \left(q_0 + (1-q_0)e^{-\mu c_X}\right) V^h\left(q(c_X), k(c_X)\right) \\ &= \left(q_0 + (1-q_0)e^{-\mu c_X}\right)\left[\frac{q(c_X)}{r} - (1-q(c_X))\frac{k(c_X)-1}{r+\gamma}\right] \\ &= \frac{q_0}{r} - \frac{1-q_0}{r+\gamma}Kc_Xe^{-\mu c_X} = \frac{q_0}{r} - \frac{1-q_0}{r+\gamma}K\min\left\{X, \frac{1}{\mu}\right\}e^{-\mu\min\{X,1/\mu\}}.\end{aligned}$$

Next, suppose $\frac{K}{\mu e} > \frac{q_0}{1-q_0}$. First, consider $X \le C_Y = 1/\mu$. By Lemma B.4, the adversarial firm's equilibrium capacity distribution is degenerate at X. Hence, the government enters hostility with posterior

$$q(X) = \frac{q_0}{q_0 + (1-q_0)e^{-\mu X}}$$

and known effective harm $k(X) = 1 + KX$.

For $X \leq c^-$, the government grants full exposure during hostility. Therefore,

$$\begin{aligned}
\Psi(X) &= \left(q_0 + (1-q_0)e^{-\mu X}\right) V^h\left(q(X), k(X)\right) \\
&= \left(q_0 + (1-q_0)e^{-\mu X}\right) \left[\frac{q(X)}{r} - (1-q(X))\frac{k(X)-1}{r+\gamma}\right] \\
&= \frac{q_0}{r} - \frac{(1-q_0)e^{-\mu X} KX}{r+\gamma}.
\end{aligned}$$

Now suppose $c^- < X \leq C_Y$. The induced capacity distribution is still degenerate at X, but the government now offers the low-posterior screening contract. Hence, from Proposition A.6,

$$\begin{aligned}
\Psi(X) &= \left(q_0 + (1-q_0)e^{-\mu X}\right) V^h\left(q(X), k(X)\right) \\
&= \left(q_0 + (1-q_0)e^{-\mu X}\right) q(X)\frac{\gamma}{r(r+\gamma)} \left(\frac{q(X)k(X)}{k(X)-1}\right)^{rk(X)/\gamma} \\
&= \frac{q_0\gamma}{r(r+\gamma)} \left(\frac{q(X)(1+KX)}{KX}\right)^{r(1+KX)/\gamma} \\
&= \frac{q_0\gamma}{r(r+\gamma)} \left(\frac{q_0(1+KX)}{\left[q_0 + (1-q_0)e^{-\mu X}\right] KX}\right)^{r(1+KX)/\gamma}.
\end{aligned}$$

Finally, consider $X > C_Y$ and set $z = \min\{X, C_H\}$. Lemma B.4 gives the adversarial firm's equilibrium distribution over capacity, and its proof characterizes the corresponding optimal hostility contract. The moving cutoff reaches z after

$$\begin{aligned}
T(z) &= \frac{\mu}{2\gamma K}\left[(1+Kz)^2 - (1+KC_Y)^2\right] \\
&= \frac{\mu}{\gamma}\left[(z - C_Y) + \frac{K}{2}(z^2 - C_Y^2)\right].
\end{aligned}$$

When $z < C_H$, the atom at z requires the hostility contract to remain on the z-indifference envelope before exposure jumps to one. Let q_z denote the posterior probability that the firm is commercial at the beginning of this interval. Then,

$$q_z = \frac{q_0}{q_0 + (1-q_0)\frac{l(z)}{M(z)}e^{-\mu z}}.$$

From the expressions for $l(z)$ and $M(z)$ in Lemma B.4,

$$\frac{l(z)}{M(z)} = \frac{(1-q_0)\frac{K}{\mu}e^{-\mu z} - q_0\left[1 + K(z - C_Y)\right]}{(1-q_0)e^{-\mu z}(1+Kz)}.$$

Substituting this expression into the posterior gives

$$q_z = \frac{q_0\mu\left(z + \frac{1}{K}\right)}{q_0 + (1-q_0)e^{-\mu z}}.$$

Therefore, the additional delay generated by the atom at z is

$$T_A(z) = \frac{1+Kz}{\gamma}\log\left(\frac{1}{q_z}\right) = \frac{1+Kz}{\gamma}\log\left(\frac{q_0 + (1-q_0)e^{-\mu z}}{q_0\mu\left(z+\frac{1}{K}\right)}\right).$$

When $z = C_H$, $q(C_H)\left(C_H + \frac{1}{K}\right) = \frac{1}{\mu}$. Thus, $T_A(C_H) = 0$.

Thus, the total time before the hostility contract grants full exposure is $D(z) = T(z) + T_A(z)$. Substituting $C_Y = 1/\mu$ gives

$$D(z) = \frac{\mu}{\gamma}\left(z - \frac{1}{\mu}\right)\left[1 + \frac{K}{2}\left(z + \frac{1}{\mu}\right)\right] + \frac{1+Kz}{\gamma}\log\left(\frac{q_0 + (1-q_0)e^{-\mu z}}{q_0\mu\left(z+\frac{1}{K}\right)}\right).$$

As we compute in the proof of Theorem 2 in Zalke (2026), the government's hostility payoff in equilibrium is

$$\Psi(X) = \frac{q_0\gamma}{r(r+\gamma)}e^{-rD(\min\{X, C_H\})}.$$

We can define $\underline{c}_X = \min\{X, C_Y\}$ and $\overline{c}_X = \min\{X, C_H\}$. Substituting these into $D(X)$ yields and let

$$D(X) = \frac{\mu}{2\gamma K}\left[(1+K\overline{c}_X)^2 - (1+K\underline{c}_X)^2\right] - \frac{1+K\overline{c}_X}{\gamma}\log\frac{w(\overline{c}_X)}{\mu\underline{c}_X}, \tag{38}$$

where $w(\cdot)$ is as defined in (22). Combining the three cases and simplifying using $\underline{c}_X$, $\overline{c}_X$ and (38),

$$\Psi(X) = \begin{cases} \dfrac{q_0}{r} - \dfrac{1-q_0}{r+\gamma}K\underline{c}_X e^{-\mu\underline{c}_X}, & X \le c^-, \\ \dfrac{q_0\gamma}{r(r+\gamma)}e^{-rD(X)}, & X \ge c^-, \end{cases}$$

where c^- is as characterised in Proposition 1, with $c^- = \infty$ when $K/\mu e \leq q_0/(1-q_0)$.

This completes the proof.

$\square$

The two lemmas prove both the parts of the proposition. $\square$

Proof of Lemma 2. Fix a peace-time exposure path $x^p = (x_t^p)_{t\geq 0}$, and let $X_t = \int_0^t x_s^p \, ds$.

First, suppose $\frac{K}{\mu e} \leq \frac{q_0}{1-q_0}$. From Lemma B.4, $H_X^* = \delta_{\min\{X,C_Y\}}$. Thus, set $C_t^* = \min\{X_t, C_Y\}$. Then $C_t^* \sim H_{X_t}^*$ and $0 \leq \dot{C}_t^* \leq x_t^p$ a.e..

Next, suppose $\frac{K}{\mu e} > \frac{q_0}{1-q_0}$.

For $C_Y < X < C_H$, define $A(c) = 1 + K(c - C_Y)$. From Proposition 2, the equilibrium distribution has lower atom, interior density, and upper atom

$$m_L(X) = \frac{1}{M(X)}, \qquad f_X(c) = \frac{K}{M(X)}, \qquad m_U(X) = 1 - \frac{A(X)}{M(X)}.$$

From calculations in the proof of *Proposition* 2, $0 < M'(X) < K$. We can then construct the following strategy:

- For $X < C_Y$, the firm chooses $\alpha = 1$.
- At $X = C_Y$, it chooses $\alpha = 0$ with probability $m_L(C_Y)$ and $\alpha = 1$ with probability $m_U(C_Y)$.
- For $C_Y < X < C_H$, for any $c \in (C_Y, X)$, its strategy is

  $$\alpha_I(X, c) = \frac{A(c)M'(X)}{KM(X)}.$$

  Because $A(c) < M(X)$ and $M'(X) < K$, this satisfies $0 < \alpha_I(X, c) < 1$.

  A firm at the lower boundary $C = C_Y$ chooses $\alpha = 0$ until it leaves that boundary at rate,

  $$\rho_L(X) = -\frac{m_L'(X)}{m_L(X)} = \frac{M'(X)}{M(X)}.$$

  Upon leaving C_Y, it follows α_I.

A firm at the upper boundary $C = X$ chooses $\alpha = 1$ until it leaves that boundary with rate

$$\rho_U(X) = -\frac{m_U'(X)}{m_U(X)} = \frac{KM(X) - A(X)M'(X)}{M(X)\big[M(X) - A(X)\big]}.$$

Upon leaving, it also follows α_I. The two switching intensities are $x_t^p \rho_L(X_t)$ and $x_t^p \rho_U(X_t)$.

- At $X = C_H$, set $\alpha = 0$.

These sequential rules generate H_X^*. Also, since every $\alpha_t \in [0, 1]$, this process satisfies $0 \le \dot{C}_t = \alpha_t x_t^p \le x_t^p$. This establishes dynamic feasibility.

Because this strategy remains in the support of H_X^* at every subsequent exposure level, it maximizes the firm's payoff at every possible future hostility date. Hostility arrival simply stops it at the current stock but it remains optimal.

Now consider any PBE. Let H_t denote the distribution over C_t induced by the adversarial firm's equilibrium strategy denoted above and let $\mathcal{C}_t^h$ denote the hostility contract that the government offers if hostility arrives at t. Let $U^g(\mathcal{C}^h)$ denote the government's hostility payoff under $\mathcal{C}^h$ conditional on the firm being commercial. For a fixed X, define

$$Z(H, \mathcal{C}^h, X) = U^F(H, \mathcal{C}^h, X) - \frac{q_0}{1 - q_0} U^g(\mathcal{C}^h),$$

where U^F follows from (4). The government's survival-weighted hostility payoff is

$$q_0 U^g(\mathcal{C}^h) - (1 - q_0) U^F(H, \mathcal{C}^h; X) = -(1 - q_0) Z(H, \mathcal{C}^h, X).$$

Thus, for a fixed distribution H, the government minimizes Z. For a fixed contract, the second term in Z does not depend on H, so the adversarial firm maximizes Z if and only if it maximizes F. The fixed-X problem can therefore be written as

$$\max_{H \in \Delta([0,X])} \min_{\mathcal{C}^h \in \mathfrak{C}^h} Z(H, \mathcal{C}^h, X).$$

For a fixed X, we show in the proof of Lemma B.4 that H_X^* is the unique equilibrium distribution. Denote the value of this problem by $v(X)$. Since the government's hostility contract is sequentially optimal in every PBE,

$$Z(H_t, \mathcal{C}_t^h, X_t) = \min_{\mathcal{C}^h \in \mathfrak{C}^h} Z(H_t, \mathcal{C}^h, X_t) \le v(X_t) = \min_{\mathcal{C}^h \in \mathfrak{C}^h} Z(H^*_{X_t}, \mathcal{C}^h, X_t) \le Z(H^*_{X_t}, \mathcal{C}_t^h, X_t).$$

Because $H^*_{X_t}$ is the unique equilibrium distribution, the first inequality is strict whenever $H_t \neq H^*_{X_t}$. Since the second term in Z does not depend on H, it follows that

$$U^F(H_t, \mathcal{C}_t^h; X_t) \le U^F(H^*_{X_t}, \mathcal{C}_t^h; X_t), \tag{39}$$

with strict inequality whenever $H_t \neq H^*_{X_t}$ for almost every t. Thus, any deviation to a feasible process for any positive measure of time such that $H_t \neq H^*_{X_t}$ is strictly suboptimal for the adversarial firm. Thus, $H_t = H^*_{X_t}$ for almost every t.

□

Proof of Proposition 3. Let $\rho = r + \lambda$. From Lemmas 2 and B.5, the government's problem can be written as

$$\max_{x_t^p \in [\phi, 1]} \int_0^\infty e^{-\rho t} \left[x_t^p + \lambda \Psi(X_t) \right] dt,$$

with $\dot{X}_t = x_t^p$. Therefore, the HJB is

$$\rho V(X) = \max_{x \in [\phi, 1]} \left\{ x + \lambda \Psi(X) + x V'(X) \right\}.$$

The maximand is linear in x. Hence, the optimal choice is $x = 1$ when $1 + V'(X) \ge 0$, and $x = \phi$ when $1 + V'(X) \le 0$. Let $G(X) = -\Psi'(X)$ whenever Ψ is differentiable. We first show that $G(X) > 0$ and $G'(X) < 0$ for $X \in [0, \overline{C})$, and that $G(X) = 0$ for $X \ge \overline{C}$.

Suppose first that $\frac{K}{\mu e} \le \frac{q_0}{1-q_0}$. From Lemma B.5, for $X < 1/\mu$,

$$G(X) = \frac{(1-q_0)K}{r+\gamma} e^{-\mu X}(1 - \mu X) > 0,$$
$$G'(X) = -\frac{(1-q_0)\mu K}{r+\gamma} e^{-\mu X}(2 - \mu X) < 0.$$

Moreover, $\Psi(X)$ is constant for $X \ge 1/\mu$. Hence, $G(X) = 0$ for $X \ge \overline{C} = 1/\mu$.

Next, suppose that $\frac{K}{\mu e} > \frac{q_0}{1-q_0}$. On the full-exposure branch, $X < c^-$, the preceding expressions continue to apply, so $G(X) > 0$ and $G'(X) < 0$.

Next, consider the testing branch, $c^- \leq X \leq 1/\mu$. Define $h(X) = X + \frac{1}{K}$, $a(X) = \frac{q(X)(1+KX)}{KX}$, and $\eta(X) = \log a(X) + h(X)\frac{d}{dX}\log a(X)$.

From Lemma B.5,

$$\frac{\Psi'(X)}{\Psi(X)} = \frac{rK}{\gamma}\eta(X),$$

where

$$\frac{d}{dX}\log a(X) = \mu[1 - q(X)] - \frac{1}{X(1+KX)}.$$

Differentiating η gives

$$\eta'(X) = \mu[1 - q(X)]\,[2 - \mu q(X)h(X)] + \frac{1}{h(X)(KX)^2} > 0.$$

The inequality follows because $a(X) \leq 1$ on the screening branch and for $X \leq 1/\mu$, $\mu q(X)h(X) = \mu X a(X) < 1$. At $X = 1/\mu$, $\eta(1/\mu) = \log a(1/\mu) + 1 - a(1/\mu) < 0$. Because η is strictly increasing, $\eta(X) < 0$ throughout the testing branch. Therefore,

$$G(X) = -\frac{rK}{\gamma}\Psi(X)\eta(X) > 0,$$
$$G'(X) = -\frac{rK}{\gamma}\Psi(X)\left[\frac{rK}{\gamma}\eta(X)^2 + \eta'(X)\right] < 0.$$

Finally, consider the mixed-capacity branch, $1/\mu < X < C_H$. Define

$$\chi(X) = \mu q(X)\left(X + \frac{1}{K}\right).$$

By the definition of C_H, $\chi(C_H) = 1$. Moreover,

$$\frac{\chi'(X)}{\chi(X)} = \mu[1 - q(X)] + \frac{1}{X + 1/K} > 0.$$

Hence, $0 < \chi(X) < 1$ for $X < C_H$. Differentiating the expression for D in Lemma B.5 gives

$$D'(X) = \frac{K}{\gamma}\left[\chi(X) - 1 - \log\chi(X)\right] > 0,$$
$$D''(X) = \frac{K}{\gamma}\left[1 - \frac{1}{\chi(X)}\right]\chi'(X) < 0.$$

Because $\Psi(X) = \frac{q_0\gamma}{r(r+\gamma)} e^{-rD(X)}$ on this branch, $G(X) = rD'(X)\Psi(X) > 0$, and its derivative $G'(X) = r\Psi(X)\left[D''(X) - r(D'(X))^2\right] < 0$.

The expressions for G is equal at the boundaries of the three regions. At $X = c^-$, the defining equation for c^- gives $G(c^-) = \frac{q_0(1-\mu c^-)}{(r+\gamma)c^-}$.

At $X = 1/\mu$, $a(1/\mu) = \chi(1/\mu)$ and

$$\eta(1/\mu) = \log \chi(1/\mu) + 1 - \chi(1/\mu) = -\frac{\gamma}{K} D'(1/\mu).$$

Hence, the expressions for G on the pure and mixed-capacity branches coincide at $X = 1/\mu$. Finally, $D'(C_H) = 0$, so $G(C_H) = 0$. Therefore, G is positive and strictly decreasing on $[0, \overline{C})$ and equals zero for $X \geq \overline{C}$. Because Ψ is constant for $X \geq \overline{C}$, the government chooses $x = 1$ once the state reaches $\overline{C}$. Thus,

$$V(\overline{C}) = \frac{1 + \lambda\Psi(\overline{C})}{\rho}.$$

Consider first the policy that chooses $x = 1$ for all $X \in [0, \overline{C}]$. Its value is

$$V_1(X) = \int_0^{\overline{C}-X} e^{-\rho t}\left[1 + \lambda\Psi(X+t)\right] dt + e^{-\rho(\overline{C}-X)} V(\overline{C}).$$

Integrating by parts gives

$$V_1'(X) = -\lambda \int_X^{\overline{C}} e^{-\rho(s-X)} G(s)\, ds.$$

Define

$$M(X) = \lambda \int_X^{\overline{C}} e^{-\rho(s-X)} G(s)\, ds.$$

Then, $1 + V_1'(X) = 1 - M(X)$. Moreover,

$$M'(X) = \lambda\left[\rho \int_X^{\overline{C}} e^{-\rho(s-X)} G(s)\, ds - G(X)\right].$$

Since G is positive and strictly decreasing, for every $X < \overline{C}$,

$$\rho \int_X^{\overline{C}} e^{-\rho(s-X)} G(s)\, ds < \rho G(X) \int_X^{\overline{C}} e^{-\rho(s-X)} ds < G(X).$$

Therefore, $M'(X) < 0$ for every $X < \overline{C}$, and $M(\overline{C}) = 0$.

First, suppose that $M(0) \leq 1$. Since M is strictly decreasing, $1 + V_1'(X) = 1 - M(X) \geq 0$ for every $X \in [0, \overline{C}]$. Thus, $x = 1$ satisfies the HJB everywhere and $\widetilde{\tau}^* = 0$.

Next, suppose that $M(0) > 1$. Because M is continuous and strictly decreasing, and $M(\overline{C}) = 0$, there exists a unique $X^* \in (0, \overline{C})$ such that

$$1 = \lambda \int_{X^*}^{\overline{C}} e^{-\rho(s-X^*)} G(s)\, ds. \tag{40}$$

Consider the policy

$$x^*(X) = \begin{cases} \phi, & X < X^*, \\ 1, & X \geq X^*. \end{cases}$$

For $X \geq X^*$, let $V(X) = V_1(X)$. Since $M(X) \leq M(X^*) = 1$,

$$1 + V'(X) = 1 - M(X) \geq 0.$$

Therefore, $x = 1$ is optimal for every $X \geq X^*$.

For $X < X^*$, the value generated by the candidate policy is

$$V(X) = \int_0^{(X^*-X)/\phi} e^{-\rho t} \left[\phi + \lambda \Psi(X + \phi t)\right] dt + e^{-\rho(X^*-X)/\phi} V(X^*).$$

By construction,

$$\rho V(X) = \phi + \lambda \Psi(X) + \phi V'(X).$$

Let $m(X) = 1 + V'(X)$. Differentiating this gives

$$\phi m'(X) - \rho m(X) = \lambda G(X) - \rho,$$

with $m(X^*) = 0$. Solving backward from X^* gives

$$m(X) = 1 - e^{-\rho(X^*-X)/\phi} - \frac{\lambda}{\phi}\int_X^{X^*} e^{-\rho(s-X)/\phi} G(s)\, ds.$$

Because G is strictly decreasing, for every $X < X^*$,

$$\frac{\lambda}{\phi}\int_X^{X^*} e^{-\rho(s-X)/\phi} G(s)\, ds > \frac{\lambda G(X^*)}{\rho}\left[1 - e^{-\rho(X^*-X)/\phi}\right].$$

Moreover, (40) and the strict decline of G imply

$$1 = \lambda\int_{X^*}^{\overline{C}} e^{-\rho(s-X^*)} G(s)\, ds < \lambda G(X^*)\int_{X^*}^{\infty} e^{-\rho(s-X^*)} ds = \frac{\lambda G(X^*)}{\rho}.$$

Therefore, $m(X) < \left[1 - e^{-\rho(X^*-X)/\phi}\right]\left[1 - \frac{\lambda G(X^*)}{\rho}\right] < 0$. Thus, $x = \phi$ is optimal for every $X < X^*$, while $x = 1$ is optimal for every $X > X^*$. At $X = X^*$, the government is indifferent between all $x \in [\phi, 1]$. Therefore, this policy satisfies the HJB everywhere.

While the government chooses minimal exposure, the state evolves according to $\dot{X}_t = \phi$. Hence, $\widetilde{\tau}^* = \frac{X^*}{\phi}$. Substituting $X^* = \phi\widetilde{\tau}^*$ into (40) gives

$$1 = \lambda\int_{\phi\widetilde{\tau}^*}^{\overline{C}} e^{-(r+\lambda)(s-\phi\widetilde{\tau}^*)}\left[-\Psi'(s)\right] ds.$$

Finally,

$$M(0) = \lambda\int_0^{\overline{C}} e^{-(r+\lambda)s}\left[-\Psi'(s)\right] ds.$$

Therefore, if $M(0) \leq 1$, the government chooses full exposure from the beginning and $\widetilde{\tau}^* = 0$.

□

Proof of Lemma 3. Fix X and let $x \in \mathcal{X}$ satisfy $\int_0^T x_t\, dt = X$. Denote the cumulative exposure at t as

$$X(t) := \int_0^t x_s\, ds, \qquad t \in [0, T].$$

Then, integrating by parts

$$\int_0^T e^{-rt} x_t\, dt = e^{-rT_h} X(T) + r\int_0^T e^{-rt} X(t)\, dt = e^{-rT} X + r\int_0^T e^{-rt} X(t)\, dt.$$

Since $e^{-rT}X$ is fixed, maximizing $\int_0^T e^{-rt}x_t\,dt$ is equivalent to maximizing $\int_0^T e^{-rt}X(t)\,dt$.

Next, for any bang–bang strategy, $X^B(t)$, $X^B(t) = t$ for $t \in [0, \tau(X))$ and $X^B(t) = \tau(X) + \phi(t - \tau(X))$ for $t \in [\tau(X), T)$. Then, $\tau(X)$ is the unique time such that the $X(T) = X$, i.e., $\tau(X) = \frac{X-\phi T}{1-\phi}$. Because $e^{-rt} > 0$, it follows that for any $X(t)$,

$$\int_0^T e^{-rt}X(t)\,dt \;\le\; \int_0^T e^{-rt}X^B(t)\,dt,$$

with strict inequality unless $X(t) = X^B(t)$ a.e. Therefore $x^B(X)$ uniquely maximizes $\int_0^T e^{-rt}x_t\,dt$.

$\square$

Proof of Proposition 4. Fix the cumulative exposure at X_T when risk begin at time T. From Lemma 2, every PBE induces $H_t = H^*_{X_t}$ for $t > T$. Feasible capacity processes are continuous in their distribution, and X_t and H^*_X are continuous in t and X respectively. Therefore, at $t = T$, $H_T = H^*_{X_T}$.

Because hostility cannot arrive before T, all pre-exposure paths that produce the same X_T, generate the same continuation payoff for the government. Denote the continuation payoff for the government at T by $V(X_T)$. Discounting results in the government front loading X_T so, conditional on X_T, the pre-T exposure policy is

$$x_t = \begin{cases} 1, & t < \tau(X_T), \\ \phi, & \tau(X_T) < t < T, \end{cases} \qquad \tau(X_T) = \frac{X_T - \phi T}{1-\phi}. \tag{41}$$

Thus, pre-T, the government's problem is to choose $X_T \in [\phi T, T]$ to maximize

$$W_A(X_T, T) = \phi\frac{1-e^{-rT}}{r} + (1-\phi)\frac{1-e^{-r\tau(X_T)}}{r} + e^{-rT}V(X_T). \tag{42}$$

Denote the optimal cumulative exposure at T by X^*_T. By the FOC, an interior optimal $X^I_T \in (\phi T, T)$ satisfies

$$\begin{aligned} 0 &= e^{-r\tau(X^I_T)} + e^{-rT}V'(X^I_T) \\ V'(X^I_T) &= -\exp\left(\frac{r(T - X^I_T)}{1-\phi}\right). \end{aligned} \tag{43}$$

If $X^*_T = \phi T$, then it must be that $-V'(\phi T) \ge e^{rT}$, and $X^*_T = T$ implies $-V'(T) \le 1$.

From equation (13), the post-T HJB is

$$(r+\lambda)V(X) = \max_{x\in[\phi,1]} \left\{x + \lambda\Psi(X) + xV'(X)\right\}.$$

Let $\widetilde{X} = \phi\widetilde{\tau}^*$, where $\widetilde{\tau}^*$ is characterized in Proposition 3. Then, from Proposition 3, the post-T policy is,

$$x^*(X) = \begin{cases} \phi, & X < \widetilde{X}, \\ 1, & X \geq \widetilde{X}. \end{cases} \tag{44}$$

Equations (41) and (44) show that there can be at most two cutoffs τ_1 and τ_2 such that the government's optimal exposure goes from 1 to ϕ at τ_1 and then back to 1 at τ_2.

Next, we characterize the cutoffs. Let $Q(X) = -V'(X)$ and recall that $G(X) = -\Psi'(X)$. Because $x^*(X) = 1$ (from (44)), the HJB becomes

$$(r+\lambda)V(X) = 1 + \lambda\Psi(X) + V'(X).$$

Differentitating it gives,

$$Q'(X) = (r+\lambda)Q(X) - \lambda G(X).$$

Since Ψ is constant after $\overline{C}$, the continuation value is constant there, so $Q(\overline{C}) = 0$. Solving backward yields,

$$Q(X) = \lambda \int_X^{\overline{C}} e^{-(r+\lambda)(s-X)} G(s)\, ds, \tag{45}$$

for $X \geq \widetilde{X}$. At $X = \widetilde{X}$, this gives

$$Q(\widetilde{X}) = \lambda \int_{\widetilde{X}}^{\overline{C}} e^{-(r+\lambda)(s-\widetilde{X})} G(s)\, ds = 1,$$

which is the cutoff equation from Proposition 3.

For $X < \widetilde{X}$, the government chooses ϕ, so the HJB is

$$(r+\lambda)V(X) = \phi + \lambda\Psi(X) + \phi V'(X).$$

Again, differentiating it yields the ODE,

$$\phi Q'(X) = (r+\lambda)Q(X) - \lambda G(X). \tag{46}$$

Solving backwards from $Q(\widetilde{X}) = 1$ yields

$$Q(X) = e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X)} + \frac{\lambda}{\phi}\int_X^{\widetilde{X}} e^{-\frac{r+\lambda}{\phi}(s-X)} G(s)\, ds. \tag{47}$$

Now, $Q(X) < 1$ for $X > \widetilde{X}$. Thus, from equations (47) and (43), an interior solution X_T^I must be $X_T^I < \widetilde{X}$, and satisfies

$$e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X_T^I)} + \frac{\lambda}{\phi}\int_{X_T^I}^{\widetilde{X}} e^{-\frac{(r+\lambda)}{\phi}(s-X_T^I)} G(s)\, ds = \exp\left(\frac{r(T - X_T^I)}{1-\phi}\right). \tag{48}$$

Thus, given an optimal X_T^*, $\tau_1^* = \frac{X_T^* - \phi T}{1-\phi}$, and the second switching date is

$$\tau_2^* = \begin{cases} T + \frac{\widetilde{X} - X_T^*}{\phi}, & X_T^* < \widetilde{X}, \\ T, & X_T^* \geq \widetilde{X}. \end{cases} \tag{49}$$

From (43), $T = X + \frac{1-\phi}{r}\log Q(X)$ and the SOC yields $1 + \frac{1-\phi}{r}\frac{Q'(X)}{Q(X)} > 0$. Define $T_I(X) := X + \frac{1-\phi}{r}\log Q(X)$ for $X < \widetilde{X}$. Differentiating $T_I(X)$ yields $T_I'(X) = 1 + \frac{1-\phi}{r}\frac{Q'(X)}{Q(X)}$. Thus, any interior local maximizer must satisfy $T = T_I(X)$ and $T_I'(X) > 0$. From (46),

$$\begin{aligned} \phi Q'(X) &= (r+\lambda)Q(X) - \lambda G(X) \\ \frac{r\phi Q(X)}{1-\phi}(T_I'(X) - 1) &= (r+\lambda)Q(X) - \lambda G(X) \\ T_I'(X) &= \frac{(r + \lambda(1-\phi))Q(X) - \lambda(1-\phi)G(X)}{r\phi Q(X)}. \end{aligned}$$

Thus, $T_I'(X) > 0 \iff E(X) := (r + \lambda(1-\phi))Q(X) - \lambda(1-\phi)G(X) > 0$. Differentiating E along with the ODE for Q gives

$$E'(X) - \frac{r+\lambda}{\phi}E(X) = -\lambda[rG(X) + (1-\phi)G'(X)].$$

This is the ODE

$$\frac{d}{dX}\left[e^{-\frac{r+\lambda}{\phi}X}E(X)\right] = -\lambda e^{-\frac{r+\lambda}{\phi}X}\left[rG(X) + (1-\phi)G'(X)\right]$$

We first show that Q is strictly decreasing on $[0, \widetilde{X}]$. Define $J(X) := \lambda G(X) - (r+\lambda)Q(X)$.

From (46), $Q'(X) = -\frac{J(X)}{\phi}$. Differentiating J and using (46) gives $J'(X) - \frac{r+\lambda}{\phi}J(X) = \lambda G'(X) < 0$. Moreover,

$$1 = \lambda \int_{\widetilde{X}}^{\overline{C}} e^{-(r+\lambda)(s-\widetilde{X})} G(s)\, ds < \frac{\lambda G(\widetilde{X})}{r+\lambda},$$

where the inequality follows because G is strictly decreasing. Hence, $J(\widetilde{X}) = \lambda G(\widetilde{X}) - (r+\lambda) > 0$. Solving the ODE for J backward from $\widetilde{X}$ yields $J(X) > e^{-\frac{r+\lambda}{\phi}(\widetilde{X}-X)} J(\widetilde{X}) > 0$, for every $X < \widetilde{X}$. Therefore, $Q'(X) < 0$, $Q(X) > 1$, and $G(X) > \frac{r+\lambda}{\lambda}Q(X) > \frac{r+\lambda}{\lambda}$ for every $X < \widetilde{X}$.

We next show that there exists at most one interior local maximizer. The cutoff equation implies $G(\widetilde{X}) > 1 + \frac{r}{\lambda}$. Hence, $1 - \frac{r}{\lambda[G(\widetilde{X})-1]} > 0$. For $\phi < 1 - \frac{r}{\lambda[G(\widetilde{X})-1]}$, we have

$$E(\widetilde{X}) = r + \lambda(1-\phi) - \lambda(1-\phi)G(\widetilde{X}) = r - \lambda(1-\phi)[G(\widetilde{X}) - 1] < 0.$$

Next, consider the term on the right-hand side of the ODE for E. On the initial branch,

$$G(X) = \frac{(1-q_0)K}{r+\gamma} e^{-\mu X}(1-\mu X).$$

Thus,

$$\frac{rG(X) + (1-\phi)G'(X)}{G(X)} = r - (1-\phi)\mu \frac{2-\mu X}{1-\mu X}. \tag{50}$$

The right-hand side of (50) is strictly decreasing in X. Hence, on the initial branch, $rG(X) + (1-\phi)G'(X)$ can change sign at most once, and only from positive to negative.

On either of the later branches, the calculations in the proof of Proposition 3 imply that $G(X) = \Psi(X)h(X)$, where $h(X) > 0$, $h'(X) < 0$, and $\Psi(X) \leq \frac{q_0\gamma}{r(r+\gamma)}$. Because Q is decreasing,

$$\frac{G'(X)}{G(X)} = \frac{\Psi'(X)}{\Psi(X)} + \frac{h'(X)}{h(X)} = -\frac{G(X)}{\Psi(X)} + \frac{h'(X)}{h(X)} < -\frac{r+\lambda}{\lambda\,[q_0\gamma/(r(r+\gamma))]}.$$

Therefore, $rG(X) + (1-\phi)G'(X) < 0$ on the later branch whenever $\phi < 1 - \frac{q_0\gamma\lambda}{(r+\gamma)(r+\lambda)}$. Thus, under the sufficiently small ϕ assumption, $rG(X) + (1-\phi)G'(X)$ changes sign at most once on $[0, \widetilde{X}]$, and only from positive to negative.

It follows from the ODE for E that $e^{-\frac{r+\lambda}{\phi}X}E(X)$ first decreases and then increases at most once. Since $E(\widetilde{X}) < 0$, its increasing terminal portion remains negative. Hence, $\{X \in [0, \widetilde{X}] : E(X) > 0\}$ is either empty or an initial interval.

Next, for the government's objective, define $R_T(X) := e^{-\frac{r(T-X)}{1-\phi}} Q(X)$. Then,

$$\frac{\partial W_A}{\partial X}(X,T) = e^{-r\tau(X)}[1 - R_T(X)],$$
$$\frac{R'_T(X)}{R_T(X)} = \frac{E(X)}{\phi(1-\phi)Q(X)}.$$

Since $E(X) > 0$ only on an initial interval, R_T first increases and then decreases. Therefore, $R_T(X) = 1$ can have at most one upward-crossing solution. At an upward crossing, $\frac{\partial W_A}{\partial X}$ can go from positive to negative, so the critical point is a local maximizer. Any other solution is a downward crossing and is a local minimizer. Thus, there exists at most one interior local maximizer. Denote it, if it exists and is feasible, by X_T^I. Equivalently, X_T^I is the unique local maximizer satisfying $T = T_I(X_T^I)$, where $T_I(X) = X + \frac{1-\phi}{r} \log Q(X)$.

When the maxima is local, $E(X) > 0$, and therefore, $T'_I(X) = \frac{E(X)}{r\phi Q(X)} > 0$. Moreover, since $Q'(X) < 0$, $T'_I(X) = 1 + \frac{1-\phi}{r}\frac{Q'(X)}{Q(X)} < 1$. Differentiating it implicitly

$$(X_T^I)'(T) = \frac{1}{T'_I(X_T^I)} > 1.$$

Consequently, both $X_T^I - \phi T$ and $X_T^I - T$ are strictly increasing. Hence, the set of T for which X_T^I is feasible is an interval. Therefore, the only candidates for the global optimum are $X_L(T) = \phi T$, $X_I(T) = X_T^I$, and $X_L(T) = T$. Let

$$W_L(T) = W_A(\phi T, T), \qquad W_I(T) = W_A(X_I(T), T), \qquad W_F(T) = W_A(T, T).$$

First, compare the interior candidate with minimal exposure. Since $\tau(\phi T) = 0$,

$$D_{IL}(T) := W_I(T) - W_L(T) = (1-\phi)\frac{1 - e^{-r\tau(X_I(T))}}{r} - e^{-rT}\int_{\phi T}^{X_I(T)} Q(s)\, ds.$$

Differentiating and using the first-order condition for $X_I(T)$ gives

$$D'_{IL}(T) = e^{-rT}\left[r\int_{\phi T}^{X_I(T)} Q(s)\, ds + \phi\left\{Q(\phi T) - Q(X_I(T))\right\}\right] > 0.$$

Hence, the interior candidate can overtake minimal exposure at most once. Denote this crossing, if it exists, by T_{IL}^A.

Next, compare full exposure with minimal exposure:

$$D_{FL}(T) := W_F(T) - W_L(T) = (1-\phi)\frac{1-e^{-rT}}{r} - e^{-rT}\int_{\phi T}^{T} Q(s)\,ds.$$

The equality $D_{FL}(T) = 0$ is equivalent to

$$1 = \frac{rT}{e^{rT}-1}\frac{1}{1-\phi}\int_{\phi}^{1} Q(uT)\,du. \tag{51}$$

The right-hand side of (51) is strictly decreasing in T. It converges to $Q(0) > 1$ as T goes to 0 and to zero as $T \to \infty$. Hence, there exists a unique $T_{FL}^A > 0$ such that $W_F(T_{FL}^A) = W_L(T_{FL}^A)$.

Finally, compare full exposure with the interior candidate. Define $\widetilde{D}_{FI}(T) := e^{rT}[W_F(T) - W_I(T)]$. Then,

$$\widetilde{D}_{FI}(T) = (1-\phi)\frac{e^{r(T-\tau(X_I(T)))}-1}{r} - \int_{X_I(T)}^{T} Q(s)\,ds.$$

Differentiating and using $e^{r(T-\tau(X_I(T)))} = Q(X_I(T))$ gives $\widetilde{D}_{FI}'(T) = Q(X_I(T)) - Q(T) > 0$. Thus, full exposure can overtake the interior candidate at most once. Denote this crossing, if it exists, by T_{FI}^A.

Define $\underline{T}_A = \min\{T_{IL}^A, T_{FL}^A\}$, and $\overline{T}_A = \max\{T_{FL}^A, T_{FI}^A\}$. If the interior candidate does not exist, set $\underline{T}_A = \overline{T}_A = T_{FL}^A$. The preceding single-crossing results imply that the selected cumulative exposure at T is

$$X_T^* = \begin{cases} \phi T, & T < \underline{T}_A, \\ X_I(T), & T \in [\underline{T}_A, \overline{T}_A), \\ T, & T \geq \overline{T}_A. \end{cases} \tag{52}$$

At either equality, the higher-exposure tie-breaking rule selects the higher cumulative-exposure candidate.

Conditional on X_T^*, the pre-T policy in (41) switches from full to minimal exposure at $\tau_1^* = \frac{X_T^* - \phi T}{1-\phi}$.

After T, the government follows the persistent-risk policy in (44). Hence, the second switching date is

$$\tau_2^* = T + \frac{1}{\phi}\max\left\{0, \widetilde{X} - X_T^*\right\}.$$

If $X_T^* = \phi T$, then $\tau_1^* = 0$. If $X_T^* = X_I(T)$, then $Q(X_T^*) = \exp\left(\frac{r(T-X_T^*)}{1-\phi}\right) > 1$, which implies $X_T^* < \widetilde{X}$ and therefore $\tau_2^* > T$. Finally, if $X_T^* = T$, optimality at the upper boundary requires $\frac{\partial W_A}{\partial X}(T,T) = e^{-rT}[1 - Q(T)] \geq 0$. Hence, $Q(T) \leq 1$, which implies $T \geq \widetilde{X}$ and therefore $\tau_2^* = T$.

Consequently, $0 \leq \tau_1^* \leq T \leq \tau_2^*$, and the optimal exposure path is

$$x_t^* = \begin{cases} 1, & t < \tau_1^*, \\ \phi, & \tau_1^* \leq t < \tau_2^*, \\ 1, & t \geq \tau_2^*. \end{cases}$$

Hence, proven.

□

Proof of Proposition 5. By Lemma 3, for any total exposure $X \in [\phi T_h, T_h]$, the government implements X using the front-loaded bang–bang path $x^B(X)$. Therefore, its reduced objective is

$$W(X, T_h) = \phi\frac{1 - e^{-rT_h}}{r} + (1-\phi)\frac{1 - e^{-r\tau(X)}}{r} + e^{-rT_h}\Psi(X),$$

where $\tau(X) = \frac{X - \phi T_h}{1-\phi}$.

Let $G(X) = -\Psi'(X)$. As shown in the proof of Proposition 3, G is continuous, strictly positive and strictly decreasing on $[0, \overline{C})$, and equals zero for $X \geq \overline{C}$. Differentiating the government's objective gives

$$\frac{\partial W}{\partial X}(X, T_h) = e^{-r\tau(X)} - e^{-rT_h}G(X). \tag{53}$$

We first consider the case $G(0) = \frac{(1-q_0)K}{r+\gamma} \leq 1$. Because G is decreasing, $G(X) \leq 1$ for every $X \geq 0$. Moreover, $X \leq T_h$ implies $\tau(X) \leq T_h$. Therefore,

$$e^{-r\tau(X)} \geq e^{-rT_h} \geq e^{-rT_h}G(X).$$

Hence, $\frac{\partial W}{\partial X}(X, T_h) \geq 0$ throughout the feasible interval. Under the higher-exposure tie-breaking rule, $X^*(T_h) = T_h$ for every T_h. This case can be represented by setting $\underline{T}' = \overline{T}' = 0$.

For the remainder of the proof, suppose that $G(0) > 1$. Define

$$\widehat{c} = \begin{cases} 1/\mu, & \dfrac{K}{\mu e} \leq \dfrac{q_0}{1-q_0}, \\ c^-, & \dfrac{K}{\mu e} > \dfrac{q_0}{1-q_0}. \end{cases}$$

On the initial branch, $X \leq \widehat{c}$, Lemma B.5 gives $G(X) = \frac{(1-q_0)K}{r+\gamma} e^{-\mu X}(1-\mu X)$. Hence, an interior critical point satisfies

$$\frac{(1-q_0)K}{r+\gamma} e^{-\mu X}(1-\mu X) = \exp\left(\frac{r(T_h - X)}{1-\phi}\right).$$

Equivalently,

$$\frac{1}{\mu} - X = \frac{r+\gamma}{(1-q_0)\mu K} \exp\left(\frac{rT_h}{1-\phi} + \left[\mu - \frac{r}{1-\phi}\right] X\right). \tag{54}$$

To characterize the solutions of (54), let $R(T_h, X) = \exp\left(-\frac{r(T_h - X)}{1-\phi}\right) G(X)$. Equation (53) can then be written as $\frac{\partial W}{\partial X}(X, T_h) = e^{-r\tau(X)}[1 - R(T_h, X)]$.

On the initial branch,

$$\frac{\partial}{\partial X} \log R(T_h, X) = \frac{r}{1-\phi} - \mu - \frac{1}{1/\mu - X},$$
$$\frac{\partial^2}{\partial X^2} \log R(T_h, X) = -\frac{1}{(1/\mu - X)^2} < 0.$$

Therefore, $\log R(T_h, X)$ is strictly concave in X, so the set $\{X : R(T_h, X) \geq 1\}$ is an interval. Hence, the $R(T_h, X) = 1$ has at most one upward-crossing solution. Because $\frac{\partial W}{\partial X}(X, T_h) = e^{-r\tau(X)}[1 - R(T_h, X)]$, this upward-crossing solution is the unique local maximum. Any second solution is a downward crossing and therefore a local minimum.

Let $X^I(T_h)$ denote this unique upward-crossing solution of (54), whenever it belongs to $[\phi T_h, \min\{T_h, \widehat{c}\}]$. Such a solution can exist only if $\frac{r}{1-\phi} > 2\mu$. If this inequality does not hold, there is no interior maximum on the initial branch.

In the no-screening case, Ψ becomes constant after $1/\mu$, and thus, $\frac{\partial W}{\partial X}(X, T_h) = e^{-r\tau(X)} > 0$ for $X > 1/\mu$. Hence, there can be no other interior maximizer.

Next, suppose that $\frac{K}{\mu e} > \frac{q_0}{1-q_0}$. We show that there can be no interior maximizer on the screening and mixed-capacity branches, $X \in [c^-, C_H]$. Let $c_0 = \frac{q_0\gamma}{r(r+\gamma)}$. The calculations in the proof of Proposition 3 show that, on these branches, $G(X) = \Psi(X)h(X)$, where $h(X) > 0$ and $h'(X) < 0$. Moreover, $\Psi(X) \le c_0$. At any interior critical point, (53) implies $G(X) = \exp\left(\frac{r(T_h - X)}{1-\phi}\right) \ge 1$ because $X \le T_h$. Therefore,

$$\frac{G'(X)}{G(X)} = \frac{\Psi'(X)}{\Psi(X)} + \frac{h'(X)}{h(X)} = -\frac{G(X)}{\Psi(X)} + \frac{h'(X)}{h(X)} \le -\frac{1}{\Psi(X)} \le -\frac{1}{c_0}.$$

Because ϕ is sufficiently small, $\phi < 1 - \frac{q_0\gamma}{r+\gamma}$, which is equivalent to $\frac{r}{1-\phi} < \frac{1}{c_0}$. Hence, at every interior critical point on these branches, $\frac{r}{1-\phi} + \frac{G'(X)}{G(X)} < 0$. Differentiating (53) and applying the first-order condition gives

$$\frac{\partial^2 W}{\partial X^2}(X, T_h) = -\frac{r}{1-\phi}e^{-r\tau(X)} - e^{-rT_h}G'(X) = -e^{-rT_h}G(X)\left[\frac{r}{1-\phi} + \frac{G'(X)}{G(X)}\right] > 0.$$

Thus, every interior critical point on the screening and mixed-capacity branches is a strict local minimum. Because G is continuous at the boundaries of the branches, none of the boundaries creates an additional kink maximum.

It follows that the government's global optimizer must be one of $X_L(T_h) = \phi T_h$, $X_I(T_h)$, or $X_F(T_h) = T_h$, where $X_I(T_h)$ is included only when the upward-crossing interior candidate exists.

The domain on which $X^I(T_h)$ exists is connected. To see this, let $a = \frac{r}{1-\phi}$. Implicitly differentiating the first-order condition

$$G(X^I(T_h)) = \exp\left(a[T_h - X^I(T_h)]\right)$$

gives

$$\frac{dX^I(T_h)}{dT_h} = \frac{a}{a + G'(X^I(T_h))/G(X^I(T_h))} > 1.$$

The denominator is positive because $X^I(T_h)$ is the upward-crossing solution, and it is smaller than a because $G' < 0$. Therefore, both $X^I(T_h) - \phi T_h$ and $X^I(T_h) - T_h$ are strictly increasing. Hence, the interior candidate can enter and leave the feasible interval at most once.

Let $W_L(T_h) = W(\phi T_h, T_h)$, $W_I(T_h) = W(X^I(T_h), T_h)$, and $W_F(T_h) = W(T_h, T_h)$, where W_I is defined only when X^I exists.

First, compare the interior candidate with minimal exposure. Let $D_{IL}(T_h) = W_I(T_h) - W_L(T_h)$. Because $\tau(\phi T_h) = 0$,

$$D_{IL}(T_h) = (1-\phi)\frac{1 - e^{-r\tau(X^I(T_h))}}{r} - e^{-rT_h} \int_{\phi T_h}^{X^I(T_h)} G(s)\, ds.$$

Differentiating and applying the first-order condition for $X^I(T_h)$ gives

$$D'_{IL}(T_h) = e^{-rT_h} \left[r \int_{\phi T_h}^{X^I(T_h)} G(s)\, ds + \phi \left\{ G(\phi T_h) - G(X^I(T_h)) \right\} \right] > 0.$$

Hence, D_{IL} crosses zero at most once. Denote this unique crossing, if it exists, by T'_{IL}.

Next, compare full exposure with minimal exposure. Let $D_{FL}(T_h) = W_F(T_h) - W_L(T_h)$. Because $\tau(T_h) = T_h$,

$$D_{FL}(T_h) = (1-\phi)\frac{1 - e^{-rT_h}}{r} - e^{-rT_h} \int_{\phi T_h}^{T_h} G(s)\, ds.$$

The condition $D_{FL}(T_h) = 0$ is equivalent to

$$1 = \frac{rT_h}{e^{rT_h} - 1} \frac{1}{1-\phi} \int_{\phi}^{1} G(zT_h)\, dz.$$

The first term on the right-hand side is strictly decreasing in T_h. Because G is decreasing, the second term is also weakly decreasing and is strictly decreasing while the interval intersects the region where $G > 0$. Moreover, the right-hand side converges to $G(0) > 1$ as T_h converges to zero and converges to zero as T_h converges to infinity. Therefore, there exists a unique positive T'_{FL} such that

$$W_F(T'_{FL}) = W_L(T'_{FL}).$$

Finally, compare full exposure with the interior candidate. Let

$$D_{FI}(T_h) = W_F(T_h) - W_I(T_h).$$

Multiplying by e^{rT_h} preserves its sign and zeros. Define $\widetilde{D}_{FI}(T_h) = e^{rT_h} D_{FI}(T_h)$. Then,

$$\widetilde{D}_{FI}(T_h) = (1-\phi)\frac{e^{r[T_h - \tau(X^I(T_h))]} - 1}{r} - \int_{X^I(T_h)}^{T_h} G(s)\, ds.$$

Differentiating and by FOC, $\widetilde{D}'_{FI}(T_h) = G(X^I(T_h)) - G(T_h) > 0$. Hence, D_{FI} crosses zero at most once. Denote this unique crossing, if it exists, by T'_{FI}.

If $X^I(T_h)$ does not exist, the only candidates are minimal and full exposure. Therefore, $\underline{T}' = \overline{T}' = T'_{FL}$.

Suppose next that $X^I(T_h)$ exists. Consider two cases.

Case 1: $T'_{FL} \leq T'_{IL}$**.**
For $T_h < T'_{FL}$, minimal exposure dominates both full exposure and the interior candidate. At T'_{FL},

$$W_F(T'_{FL}) = W_L(T'_{FL}) \geq W_I(T'_{FL}).$$

Since $\widetilde{D}_{FI}$ is strictly increasing, the interior candidate cannot subsequently overtake full exposure. Therefore, the government moves directly from minimal to full exposure at T'_{FL}, and $\underline{T}' = \overline{T}' = T'_{FL}$.

Case 2: $T'_{IL} < T'_{FL}$**.**
For $T_h < T'_{IL}$, minimal exposure dominates both other candidates. At T'_{IL}, $W_I(T'_{IL}) = W_L(T'_{IL}) > W_F(T'_{IL})$. Hence, the interior candidate becomes optimal.

At T'_{FL}, $W_I(T'_{FL}) > W_L(T'_{FL}) = W_F(T'_{FL})$, so the interior candidate remains optimal. Because $\widetilde{D}_{FI}$ is strictly increasing and full exposure is eventually optimal, there exists a unique $T'_{FI} > T'_{FL}$ such that $W_F(T'_{FI}) = W_I(T'_{FI})$. Therefore, $\underline{T}' = T'_{IL}$, and $\overline{T}' = T'_{FI}$.

Combining the cases gives

$$X^*(T_h) = \begin{cases} \phi T_h, & T_h < \underline{T}', \\ X^I(T_h), & T_h \in [\underline{T}', \overline{T}'), \\ T_h, & T_h \geq \overline{T}', \end{cases}$$

where the middle branch is omitted whenever $X^I(T_h)$ does not exist or $\underline{T}' = \overline{T}'$. The higher-exposure tie-breaking rule determines the government's choice at the transition dates.

□

Proof of Proposition A.6. The result follows directly from Theorem 1 in Kolb and Madsen (2023) after substituting the effective harm parameter $K' = 1 + KC_{T_h}$ and the beliefs at the start of hostility as q_{T_h}. Theorem 1 (Kolb and Madsen, 2023), gives the optimal contract and the hostility continuation are calculated from the payoffs in the proof of Theorem 1. □